

CRYSTALLIZATION FROM SOLUTIONS AND MELTS

Dissolved or molten substances are recoverable in solid form by precipitation upon cooling or upon removal of the solvent or by addition of precipitating agents. For convenience a distinction is made between two kinds of processes:

1. In solution crystallization, the crystals are separated away from a solvent, often water. In the case of inorganic solids particularly, the operating temperature is far below their melting points.
2. In melt crystallization, two or more substances of comparable melting points are separated by some degree of cooling. The degree of completeness of such separations depends on the phase equilibrium relations. When the crystals must be refined to remove occluded substances, the recovered material may leave the process in molten form. Subsequently, it may be solidified as flakes or sprayed granules.

The design of crystallizers is based on knowledge of phase equilibria, solubilities, rates and amounts of nuclei generation, and rates of crystal growth. Each system is

unique in most of these respects and not often predictable. The kind of information needed for design of a continuous crystallizer is indicated by the data supplied for Example 16.1.

Although theoretical advances are being made, the current state of the art of crystallization requires pilot plant evaluation of parameters of equations and of such operating variables as

1. crystal size distribution,
2. effects of impurities and additives,
3. residence time,
4. circulation rate,
5. mixing efficiency,
6. allowable degrees of supersaturation or subcooling,
7. heat transfer characteristics,

and others peculiar to the particular kind of equipment under consideration for the full scale plant.

This chapter will discuss the main concepts associated with crystallization practice, and will describe the main types of equipment used nowadays, together with some indications of their performance and applicability.

16.1. SOLUBILITIES AND EQUILIBRIA

The variation of the solubilities of most substances with temperature is fairly regular, and usually increases with temperature. When water is the solvent, breaks may occur in solubility curves because of formation of hydrates. Figure 16.1(a) shows such breaks, and they can be also discerned in Figures 16.2(b) and (c). Unbroken lines usually are well enough represented by second degree polynomials in temperature, but the Clapeyron-type equation with only two constants, $\ln x = A + B/T$, is of good accuracy, as appears for some cases on Figure 16.1(b).

A convenient unit of solubility is the mass of solute per unit mass of solvent, or commonly g solute/100 g solvent. Interconversions with molal units and mol fractions are made readily when densities of the solutions are known.

Under quiescent conditions a concentration substantially in excess of normal solubility or a temperature lower than the normal saturation temperature can be maintained. The maximum supersaturation appears to be a fairly reproducible quantity, but is reduced or even eliminated by stirring or by the introduction of dust or seed crystals. Some data are shown in Figure 16.1(c) and in Table 16.1. They are expressed as $\Delta C = C - C_{\text{sat}}$ or as $\Delta C/C_{\text{sat}}$ or as $\Delta T = T - T_{\text{sat}}$. According to the data of Table 16.1(d), subcooling correlates roughly with the heat of solution. The increments ΔC and ΔT can be quite substantial quantities.

The several regions of varying stability are represented by Figure 16.1(d). At concentrations above or temperatures below those represented by the supersaturation line, nuclei form and crystals grow spontaneously, although the rates of these processes do depend on the depth of penetration of the unstable region. Little control can be exercised on behavior in this region. In the metastable region, growth of crystals will occur even under quiescent conditions when dust or seeds are introduced and nuclei can be generated by agitation. Behavior in the metastable region is

largely controllable so that it is the practical operating region for production of crystals of significant sizes.

Practically feasible extents of supersaturation or subcooling are fairly small and depend on the substance and the temperature. Some data appear in Table 16.2. Since the recommended values are one-half the maxima listed, they rarely are more than 2°C or so. This means that very high circulation rates through heat exchangers are needed. Thus, in the urea process of Example 16.1, the temperature rise is 2°F, and the volumetric circulation rate is about 150 times the fresh feed rate.

PHASE DIAGRAMS

Equilibria between liquid and solid phases over wide ranges of temperature are represented compactly on phase diagrams. The effect of moderate pressure on condensed phases is negligible. Aqueous systems often are complicated by the formation of hydrates, and other substances also may form intermolecular compounds. Of the substances of Figure 16.2, KCl does not form a hydrate, but NaCl and MgSO₄ do. Mixtures always have lower melting points than those of the pure components. The lowest temperature and the corresponding composition at which a liquid phase can be present identify the eutectic ("easy melting"), for example, point C on Figure 16.2(a) and point B on Figure 16.2(b). Binary and ternary eutectics also are identified on the ternary diagram [Fig. 16.2(f)].

The effects of evaporation or chilling on the amounts and compositions of the liquid and solid phases can be followed on the diagrams. Example 16.2 does this. Mixtures that form eutectics cannot be separated completely by chilling. The amount and nature of a separated solid phase depends on the temperature and the overall composition. Examples 16.2(c) and (d) make such calculations. Mixtures that are completely miscible in both liquid and solid phases, such as Figure 16.2(d), can be separated

EXAMPLE 16.1**Design of a Crystallizing Plant**

A plant is to make 10,000 lb/hr of urea crystals from a solution that contains 75% dissolved salt. The material balance and operating conditions are shown on the sketch. Key crystallization data are given by Bennett (1981, p. 452) as

1. The residence time is 3.4 hr.
2. The temperature rise across the heater is 2°F.

Other information deduced from pilot plant work is:

3. The feed contains 75% solids, but 1200 lb/hr of wash water from the centrifuge also is returned to the crystallizer.
4. The liquor contains 66.8% dissolved urea and has a specific gravity of 1.17 at the operating temperature of 130°F.
5. The slurry contains 28 lb crystals/cuft and has a specific gravity of 1.354.
6. A purge stream amounting to 7% of the feed liquor is withdrawn as shown on the sketch.
7. The pressure is 60 Torr, at which the saturation temperature of steam is 106°F. The superheat of 24°F is neglected in figuring the vapor density and velocity.
8. Depth of liquid in the vessel should not exceed 10 ft and the vapor velocity should not exceed that given by the equation, $u = 0.06\sqrt{\rho_L/\rho_g} - 1$. A wire mesh deentrainer is not feasible because of encrustation.
9. Heat capacity of the solid is 0.62 Btu/(lb)(°F) and the heat of crystallization is 104 Btu/lb.
10. For sizing the vacuum ejector, air leakage is estimated at 25 lb/hr and carbon dioxide is 34 lb/hr.
11. The coefficient of heat transfer in the exchanger is 200 Btu/(hr)(sqft)(°F).

Calculations:

$$\begin{aligned}\text{vapor rate} &= 4471(296)/3600 = 367.6 \text{ cfs,} \\ \text{slurry holdup} &= 10,000(3.4)/28 = 1214.3 \text{ cuft,} \\ u_{\max} &= 0.06\sqrt{84.5/0.0034} - 1 = 9.46 \text{ fps,} \\ D_{\min} &= \sqrt{367.6/9.46(\pi/4)} = 7.03 \text{ ft.}\end{aligned}$$

The corresponding liquid depth is

$$h = 1214.3/(\pi/4)D^2 = 31.0 \text{ ft,}$$

which is too great a value.

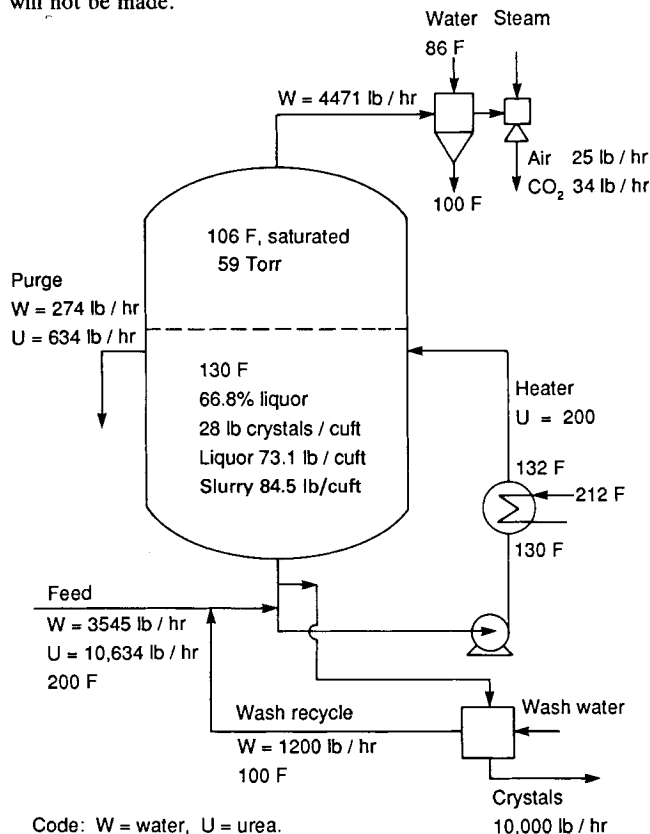
Try $D = 12.5$ ft:

dished head capacity = 152 cuft (Figure 18.5),

straight side = $(1214.3 - 152)/((12.5)^2(\pi/4)) = 8.66$ ft, say 9.0 ft.

Together with the depth of liquid in the dished head, the total depth will be close to the 10 ft specified as the maximum. From Figure 18.5, a free board of 5.5 ft is adequate in the absence of a deentraining pad. Accordingly, the vessel will have a diameter of 12.5 ft, a straight side of 14.5 ft, and dished heads designed for full vacuum. The sketch is to scale.

Sufficient data are given for finding the heat balance and the liquor circulation rate, and for sizing the auxiliaries such as lines, pump, heat exchanger and vacuum system, but those calculations will not be made.



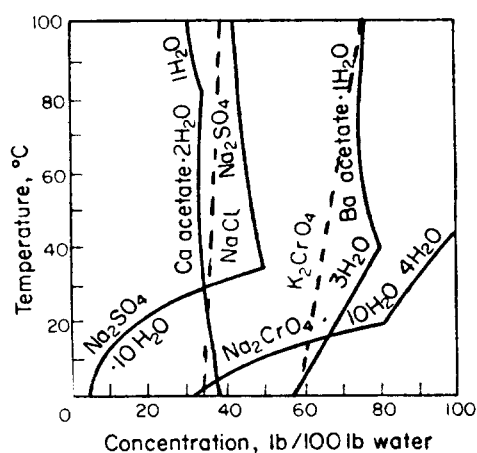
essentially completely in multistage equipment, although such processes are not often feasible. The possible extent of separation of multicomponent mixtures can be interpreted with a phase diagram like those of Figure 16.2(f) and Example 16.3. Phase diagrams are fairly plentiful, but published ones usually seem to be of the system they were interested in and not of the one you are interested in. Fortunately, nowadays phase diagrams can be developed at moderate cost and expenditure of time with differential scanning calorimeters.

Estimates of phase diagrams can be made on the assumption of ideal behavior or with activity coefficient data based on binary measurements that are more easily obtained. In such cases, clearly, it should be known that intermolecular compounds do not form.

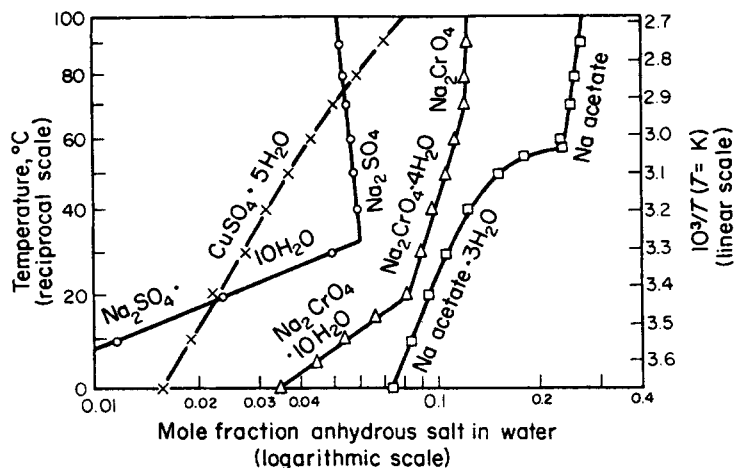
The freezing behaviors of ideal mixtures over the entire range of temperatures can be calculated readily. The method is explained for example by Walas (*Phase Equilibria in Chemical Engineering*, Butterworths, Stoneham, MA, 1985, Example 8.9).

ENTHALPY BALANCES

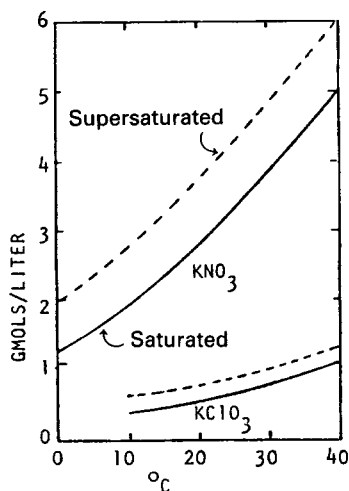
Although the thermal demands of crystallization processes are small compared with those of possibly competitive separation processes such as distillation or adsorption; nevertheless, they must be known. For some important systems, enthalpy-composition diagrams have been prepared, like those of Figure 16.3, for instance. Calculations also may be performed with the more widely



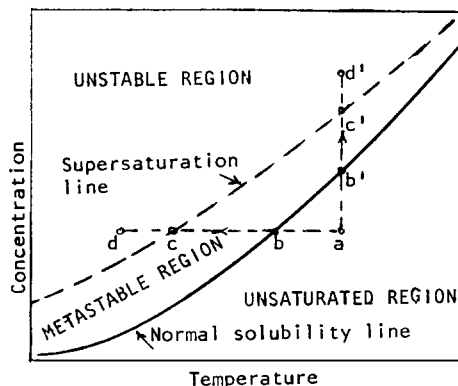
(a)



(b)



(c)



(d)

Figure 16.1. Solubility relations. (a) Linear plot of solubilities against temperature (Mullin, 1972). (b) Solubility against temperature plotted according to the equation $z = \exp(A + B/T)$ (Mullin, 1972). (c) Normal and supersolubilities of two salts (data collected by Khamiskii, 1969). (d) Identification of regions on solubility plots. In the unstable region, nucleation and growth are spontaneous. In the metastable region growth can occur on externally introduced particles. Along $a-d$ to the left or along $a-d'$ upwards, nucleation and growth can start at c or c' , but a substantial nuclei growth rate will not be achieved until d or d' are reached.

available data of heat capacities and heats of solution. The latter are most often recorded for infinite dilution, so that their utilization will result in a conservative heat balance. For the case of Example 16.3, calculations with the enthalpy-concentration diagram and with heat of solution and heat capacity data are not far apart.

16.2. CRYSTAL SIZE DISTRIBUTION

Crystal size distribution (CSD) is measured with a series of standard screens. The openings of the various mesh sizes according to the U.S. Standard are listed in Example 6.6, and according to the British Standard in Figure 16.4. Table 12.1 is a complete listing. The size of a crystal is taken to be the average of the screen openings of successive sizes that just pass and just retain the crystal.

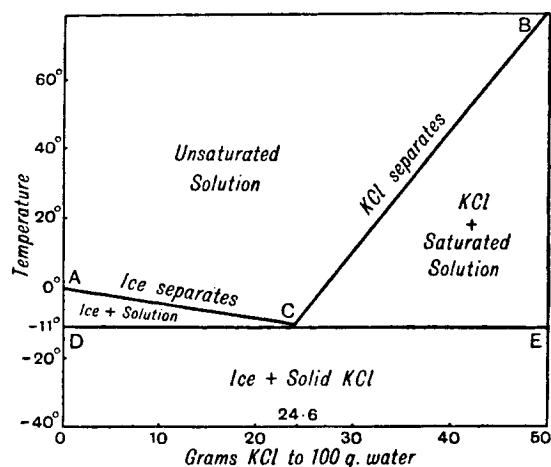
The cumulative wt % either greater or less than a specified screen opening is recorded. The amount of a size less than a

particular screen opening and greater than the next smaller size is called the differential amount. Typical size distribution data on Figure 16.4 are plotted in two cumulative modes, greater than or less than, and as differential polygons or histograms. For some purposes the polygon may be smoothed and often is shown that way. Some theoretical cumulative and differential distribution curves of similar nature are shown in Figure 16.5; the abscissas are proportional to the crystal length.

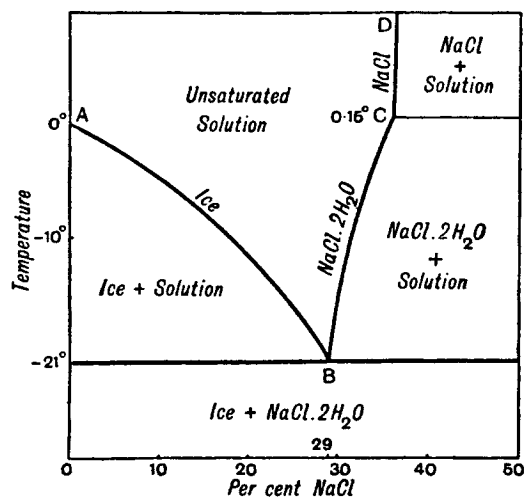
Cumulative data often are represented closely by the Rosin-Rammler-Sperling (RRS) equation

$$y = 100 \exp[-(d/d_m)^n], \quad (16.1)$$

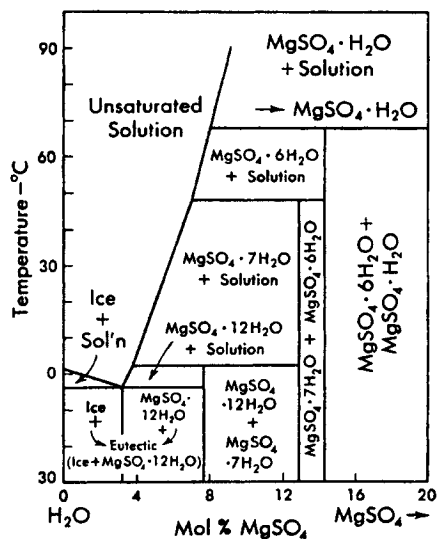
where d is the diameter, d_m is a mean diameter corresponding to $y = 100/e = 36.8\%$ and n is called the uniformity factor. The greater n , the more nearly uniform the distribution. The log-log plot of this



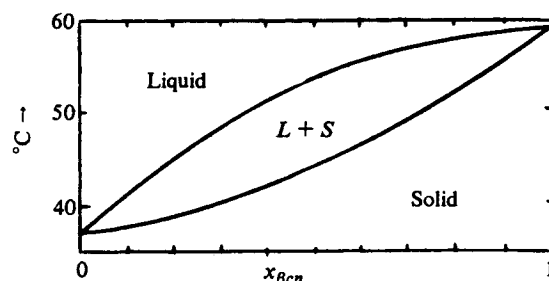
(a)



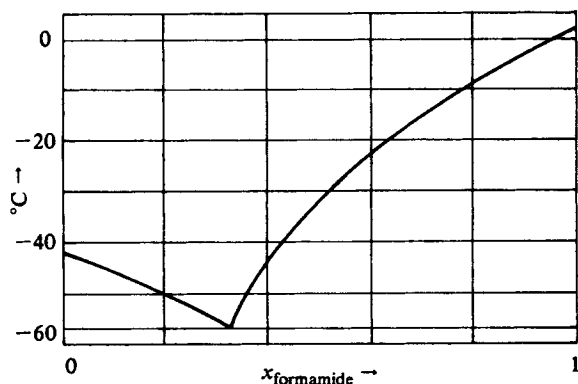
(b)



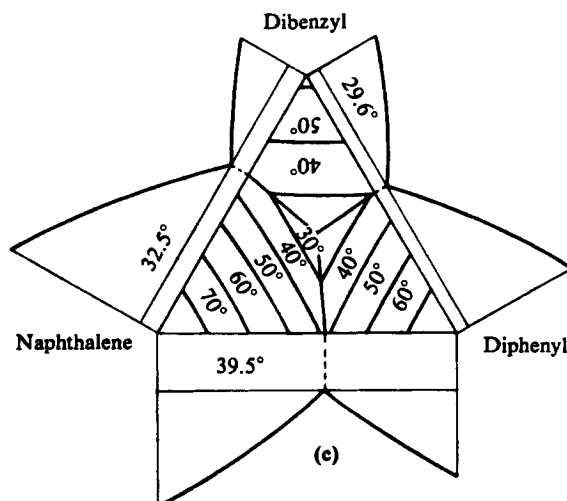
(c)



(d)



(e)



(f)

Figure 16.2. Some phase diagrams. (a) The water end of the system potassium chloride and water. (b) The water end of the system sodium chloride and water. (c) The water end of the system magnesium sulfate and water; the heptahydrate goes to the mono at 150°C, and to anhydrous at 200°C. (d) β -methylnaphthalene and β -chloronaphthalene form solid solutions. (e) Mixtures of formamide and pyridine form a simple eutectic. (f) These mixtures form binary eutectics at the indicated temperatures and a ternary eutectic at mol fractions 0.392 dibenzyl, 0.338 diphenyl, and 0.27 naphthalene.

TABLE 16.1. Data of Supersaturation and Subcooling of Solutions

(a) Maximum Supersaturation of Solutions at 20°C, $\beta = \Delta C / C_0$

Solute	Tovbin and Krasnova's data [144]	Gorbachev and Shlykov's data [34]	Fisher's data [152]
KCl	0.095	0.39	—
KBr	0.056	0.102	—
KI	0.029	—	—
KClO ₃	0.41	—	—
KNO ₃	0.36	1.08	—
NH ₄ NO ₃	0.10	—	—
NaNO ₃	0.064	—	—
Mg(NO ₃) ₂	0.93	—	—
K ₂ SO ₄	0.37	—	0.34
K ₂ C ₂ O ₄	0.41	—	—
K ₂ CrO ₄	0.093	—	—
K ₂ Cr ₂ O ₇	0.62	—	0.32
Ba(NO ₃) ₂	0.40	—	—
CuSO ₄	1.50	—	—
HgCl ₂	0.43	—	—
K ₃ Fe(CN) ₆	0.13	—	—
K ₄ Fe(CN) ₆	0.54	—	—
KBrO ₃	—	2.71	—
KIO ₃	—	1.60	—
Na ₂ C ₂ O ₄	—	—	0.86
(NH ₄) ₂ C ₂ O ₄ · H ₂ O	—	—	0.36

* Fisher's results were obtained at 25°C.

(b) Temperature Dependence of the Maximum Supersaturation of Salt Solutions

Solute	<i>t</i> , °C	<i>C</i> ₀ , moles/liter	<i>C</i> , moles/liter	$\beta = C - C_0$	β
KNO ₃	0	1.25	2.03	0.78	0.62
	10	1.96	2.78	0.81	0.41
	20	2.76	3.75	0.99	0.36
	30	3.83	4.84	1.01	0.26
	40	4.97	6.00	1.03	0.20
KCl	0	3.33	3.88	0.55	0.16
	10	3.72	4.12	0.40	0.11
	20	4.03	4.42	0.39	0.095
	30	4.29	4.45	0.16	0.037
	40	4.45	4.58	0.13	0.029
KClO ₃	10	0.40	0.65	0.25	0.62
	20	0.58	0.82	0.24	0.41
	30	0.80	1.05	0.25	0.32
	40	1.11	1.32	0.21	0.19
K ₂ CrO ₄	10	2.68	2.96	0.28	0.11
	20	2.74	3.00	0.26	0.093
	30	2.82	3.03	0.21	0.073
	40	2.98	3.07	0.19	0.065

(c) Maximum Supercooling of Salt Solutions at Various Temperatures

Solute	Heat of solution, λ , cal/mole	$t_0 - t$, °C	$\lambda(t_0 - t)$, cal-mole ⁻¹ deg ⁻¹
KCl	4046	19.6	78897
KBr	5080	16.3	80804
KI	5110	15.5	79205
KBrO ₃	9760	8.8	84788
KIO ₃	6780	13.5	91490
KClO ₃	9950	6.6	65670
KNO ₃	8800	13.0	114400
KClO ₄	12100	6.3	76230
KCNS	6100	13.0	79300
NaNO ₃	5030	13.0	65399
NaClO ₃	5600	12.0	67200
NaCl	12200	51.0	62220
NH ₄ Cl	3880	20.0	77600
(NH ₄) ₂ SO ₄	2370	24.0	56880
NH ₄ NO ₃	6320	10.3	65016
HgCl ₂	3300	25.0	82500
CuSO ₄	2750	36.7	80925
NaClO ₄	3600	20.0	72000
NH ₄ ClO ₄	6360	12.0	76320
Ba(ClO ₄) ₂	9400	9.0	84600

(d) Dependence of the Maximum Supercooling of Solutions on Heat of Solution

Solute	<i>t</i> ₀ , °C	<i>t</i> , °C	$\theta = t_0 - t$, °C	Solute	<i>t</i> ₀ , °C	<i>t</i> , °C	$\theta = t_0 - t$, °C
KNO ₃	20	-1.0	21.0	KBr	10.5	-1.8	12.3
	30	8.9	21.1		20	8.0	12.0
	40	18.9	21.1		30	17.8	12.2
	50	28.8	21.2		40	28.0	12.0
	60	38.8	21.2		50	37.9	12.1
	70	48.9	21.1		60	47.7	12.3
KCl	50	6.7	43.3	K ₂ SO ₄	80	67.7	12.3
	60	16.6	43.4		90	3.0	87.0
	70	26.7	43.3		100	13.0	87.0
	80	36.7	43.3				
	90	46.6	43.4				

(Khamskii, 1969).

of variation is defined by the equation

$$CV = 100(d_{16} - d_{84})/2d_{50}. \quad (16.2)$$

equation should be linear. On Figure 16.4(c) the scatter about the straight line is small, but several of the plots of commercial data of Figure 16.6 deviate somewhat from linearity at the larger diameters.

Two other single numbers are used to characterize size distributions. The median aperture, MA or d_{50} , is the screen opening through which 50% of the material passes. The coefficient

The origin of this concept is that the fraction of the total area under a normal distribution curve between the 16 and 84% points is twice the standard deviation. The smaller CV, the more nearly uniform the crystal sizes. Products of DTB crystallizers, for instance, often have CVs of 30–50%. The number is useful as a measure of consistency of operation of a crystallizer. Some details are given by Mullin (1972, pp. 349, 389).

TABLE 16.2. Maximum Allowable Supercooling ΔT (°C) and Corresponding Supersaturation ΔC (g/100 g water) at 25°C^a

Substance	ΔT	ΔC
NH ₄ alum	3.0	1.0
NH ₄ Cl	0.7	0.3
NH ₄ NO ₃	0.6	3.0
(NH ₄) ₂ SO ₄	1.8	0.5
NH ₄ H ₂ PO ₄	2.5	2.3
CuSO ₄ ·5H ₂ O	1.4	1.0
FeSO ₄ ·7H ₂ O	0.5	0.6
K alum	4.0	1.0
KBr	1.1	0.6
KCl	1.1	0.3
KI	0.6	0.4
KH ₂ PO ₄	9.0	4.6
KNO ₃	0.4	0.6
KNO ₂	0.8	0.8
K ₂ SO ₄	6.0	1.3
MgSO ₄ ·7H ₂ O	1.0	1.3
NiSO ₄ ·7H ₂ O	4.0	4.4
NaBr·2H ₂ O	0.9	0.9
Na ₂ CO ₃ ·10H ₂ O	0.6	2.8
Na ₂ CrO ₄ ·10H ₂ O	1.6	0
NaCl	4.0	0.2
Na ₂ B ₄ O ₇ ·10H ₂ O	4.0	0.9
NaI	1.0	1.7
NaHPO ₄ ·12H ₂ O	0.4	1.5
NaNO ₃	0.9	0.7
NaNO ₂	0.9	0.6
Na ₂ SO ₄ ·10H ₂ O	0.3	0.7
Na ₂ S ₂ O ₃ ·5H ₂ O	1.0	2.2
Urea	2.0	

^a Working values usually are not more than one-half the maxima.
(After Mullin, 1972).

16.3. THE PROCESS OF CRYSTALLIZATION

The questions of interest are how to precipitate the crystals and how to make them grow to suitable sizes and size distributions. Required sizes and size distributions are established by the need for

subsequent recovery in pure form and ease of handling, and by traditional commercial practices or consumer preferences.

CONDITIONS OF PRECIPITATION

The most common methods of precipitating a solid from a solution are by evaporation of the solvent or by changing to a temperature at which the solubility is lower. Usually solubility is decreased by lowering the temperature. Some examples are in Figure 16.1. The limit of removal is determined by the eutectic composition. According to the data of Figure 16.2, for instance, a 24.6% solution of KCl will solidify completely at -11°C and a 3.5% solution of MgSO₄ will do so at 4°C; these values represent the limits to which salt is recoverable by chilling. Complete recovery, however, is accomplished by evaporation.

A precipitate may be formed as a result of chemical reaction between separately soluble gases or liquids. Commercial examples are productions of sodium sulfate, ammonium sulfate, and ammonium phosphate.

Precipitation also can be induced by additives, a process generally called salting out because salts with ions common to those whose precipitation is desired are often used for this purpose. For instance, ammonium chloride is recovered from spent Solvay liquors by addition of sodium chloride and the solubility of BaCl₂ can be reduced from 32% to 0.1% by addition of 32% of CaCl₂. Other kinds of precipitants also are used, for instance, alcohol to precipitate aluminum sulfate from aqueous solutions.

Foreign substances even in minute amounts may have other kinds of effects on crystallization: They may inhibit or accelerate growth rate or change the shape of crystals, say from rounded to needlelike, or otherwise. One of the problems sometimes encountered with translating laboratory experience to full scale operation is that the synthetic liquors used in the laboratory may not contain the actually occurring impurities, and thus give quite different performance. Substances that modify crystal formation are very important industrially and many such materials have been the subject of patents.

SUPERSATURATION

A saturated solution is one that is in equilibrium with the solid phase and will remain unchanged indefinitely at a particular

EXAMPLE 16.2

Using the Phase Diagrams of Figure 16.2

- Evaporation of a solution of MgSO₄ at 30°C: As water is removed, the composition moves along the horizontal. When the salt concentration reaches about 6%, precipitation of heptahydrate begins and is completed at about 13%. Between 13 and 14% salt, the precipitate is a mixture of solid hepta and solid hexa hydrates. Beyond 14%, the mixture consists of mixtures of solid hepta and mono hydrates in proportions determined by the amount of water present overall.
- Chilling of a 6% solution of MgSO₄: precipitation of heptahydrate begins at about 35°C. At about 2°C, the mixture consists of solid dodecahydrate and unsaturated solution. Below -4°C complete solidification exists; the product is a mixture of pure dodecahydrate and an intimate eutectic mixture of ice and dodeca crystals.
- Recovery of pyridine: As appears on the diagram of Figure

16.2(e), the eutectic contains 33% formamide and 67% pyridine. When the mixture contains 80% pyridine, the maximum possible recovery of pure pyridine is

$$P = (0.8 - 0.67)/(1 - 0.67) = 0.39, \text{ or } 39\%.$$

- Recovery of formamide: When the mixture of Figure 16.2(e) contains 80% formamide, the maximum recovery of the pure material is

$$F = (0.8 - 0.33)/(1 - 0.33) = 0.70, \text{ or } 70\%.$$

- At 50°C, the liquid phase of Figure 16.2(d) contains 35% and the solid phase 74% of β -chloronaphthalene.
- The progress of crystallization of a ternary mixture such as that of Figure 16.2(f) is described in Example 16.8.

EXAMPLE 16.3**Heat Effect Accompanying the Cooling of a Solution of MgSO_4**

A 30% solution of MgSO_4 is cooled from 150°F to 50°F. Data of the initial and final conditions are taken off the equilibrium diagram, Fig. 16.3(b). At the lower temperature, 27% of the mixture crystallizes out as the heptahydrate.

	Original at 150°F	Final (at 50°F)		
		Total	Liquid	Solid
Water (lb)	38.6	38.6	38.6	—
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (lb)	61.4	61.4	34.4	27.0
Total (lb)	100	100	73.0	27.0
H (Btu/lb)	-3	-82	-53	-161

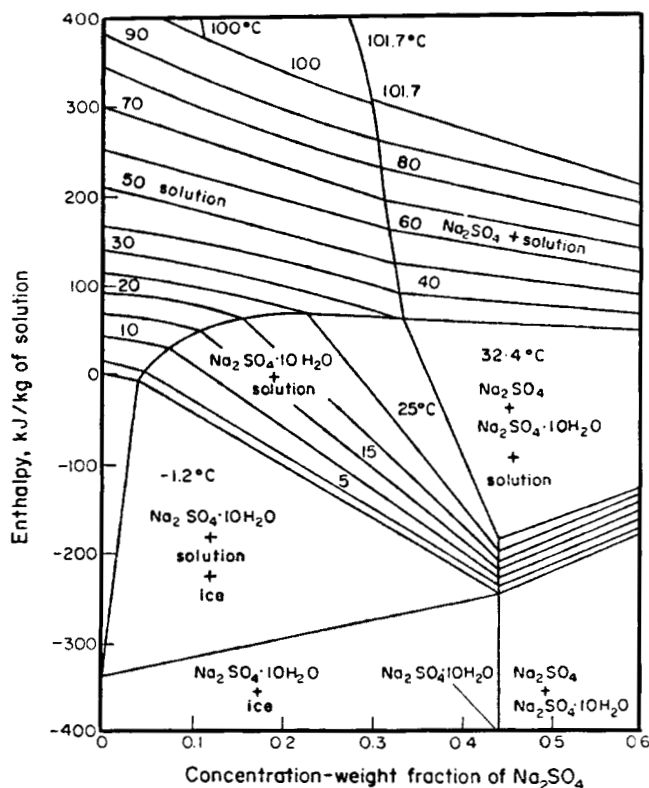
Accordingly the change in enthalpy is

$$\Delta H = -82 - (-3) = -79 \text{ Btu/lb.}$$

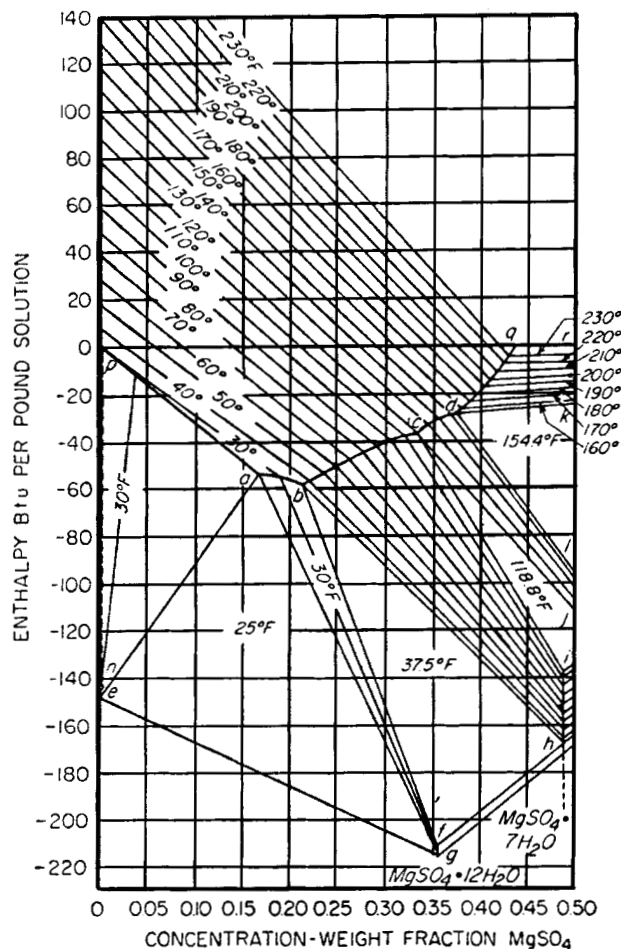
This value will be compared with a calculation using data of heat capacities and heat of solution. From *Perry's Chemical Engineers' Handbook* (1984), the heat solution of the heptahydrate is -39.2 Btu/lb and its heat capacity is $0.36 \text{ Btu/(lb)(°F)}$. The enthalpy change of the cooling and crystallization process is

$$\begin{aligned} \Delta H &= [0.386 + 0.614(0.36)](50 - 150) + 0.27(-39.2) \\ &= -71.3 \text{ Btu/lb,} \end{aligned}$$

which is a poor check of the value found with the aid of the equilibrium diagram. Possible sources of error of the second method include the use of heat of solution at infinite dilution instead of the prevailing concentration and the assumption that the heat capacities are additive.



(a)



(b)

Figure 16.3. Enthalpy-composition diagrams of some salt solutions. Several other diagrams are in the compilation of Landolt-Börnstein, IV 4b, 1972, pp. 188-224. (a) sodium sulfate/water; (b) magnesium sulfate/water (after *Chemical Engineers' Handbook*, 1963 edition, McGraw-Hill, New York); (c) sodium carbonate/water.

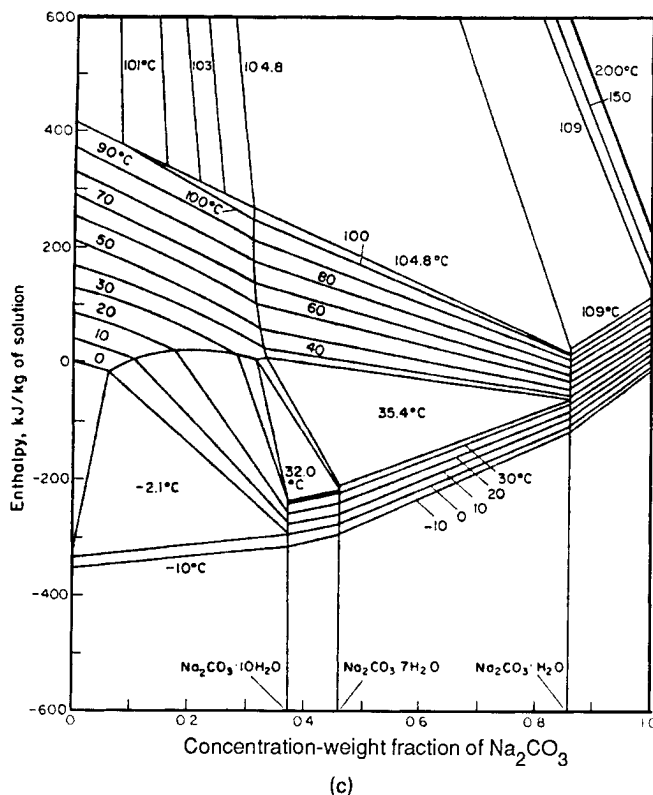


Figure 16.3—(continued)

temperature and composition of other constituents. Greater than normal concentrations also can be maintained in what is called a supersaturated condition which is metastable. Metastability is sensitive to mechanical disturbances such as agitation, ultrasonics, and friction and the introduction of solid particles. Under those

conditions, solids will separate out until normal saturation is obtained. When great care is taken, the metastable state is reproducible. A thermodynamic interpretation of metastability can be made in terms of the Gibbs energy of mixtures. In Figure 16.5(a), the solid line $a-b$ is of unsaturated solution and the straight line $b-e$ is of mixtures of all proportions of pure solid and saturated solution represented by point b . Points c and d are at the points of inflection of the plot and represent the limits of metastability. Thus line $b-c$ represents the range of concentrations between the saturated and supersaturated values.

Several measures of supersaturation are being used in terms of the saturation concentration C_0 ; thus

$$\alpha = \Delta C_s = C - C_0, \quad \text{the difference in concentrations,}$$

$$\beta = \Delta C_s / C, \quad \text{the relative difference,}$$

$$\gamma = C / C_0 = \beta + 1, \quad \text{the concentration ratio,}$$

with similar definitions for subcooling or superheating. The data of Figure 16.1(c) and Table 16.1 show that excess concentration and metastable cooling can be quite substantial amounts.

GROWTH RATES

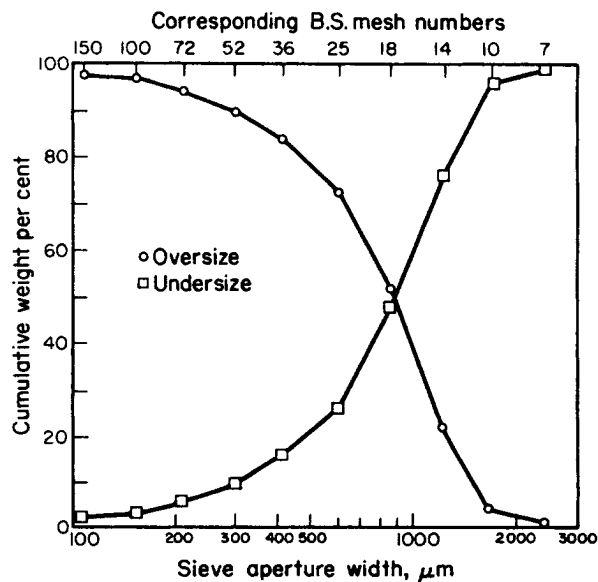
Crystallization can occur only from supersaturated solutions. Growth occurs first by formation of nuclei and then by their gradual growth. At concentrations above supersaturation, as at point d' on Figure 16.1(d), nucleation is conceived to be spontaneous and rapid. In the metastable region, nucleation is caused by mechanical shock or friction and secondary nucleation can result from the breakup of already formed crystals. It has been observed that the rate depends on the extent of supersaturation; thus

$$\frac{dC_N}{dt} = k(C - C_0)^m. \quad (16.3)$$

Values of the exponent m have been found to range from 2 to 9, but have not been correlated to be of quantitative value for prediction.

B.S. mesh number	Sieve aperture, μm	Fractional weight per cent retained	Cumulative weight per cent oversize	Cumulative weight per cent undersize
7	2360	1.2	1.2	98.8
10	1700	2.9	4.1	95.9
14	1180	18.8	22.9	77.1
18	850	28.8	51.7	48.3
25	600	22.0	73.7	26.3
36	425	11.1	84.8	15.2
52	300	6.0	90.8	9.2
72	212	3.9	94.7	5.3
100	150	1.8	96.5	3.5
150	106	1.3	97.8	2.2
> 150	—	2.2	—	—

(a)



(b)

Figure 16.4. Several ways of recording the same data of crystal size distribution (CSD) (Mullin, 1972). (a) The data. (b) Cumulative wt % retained or passed, against sieve aperture. (c) Log-log plot according to the RRS equation $P = \exp[(-d/d_m)^n]$; off this plot, $d_{50\%} = 850$, $d_m = 1000$, $n = 1.8$. (d) Differential polygon. (e) Differential histogram.

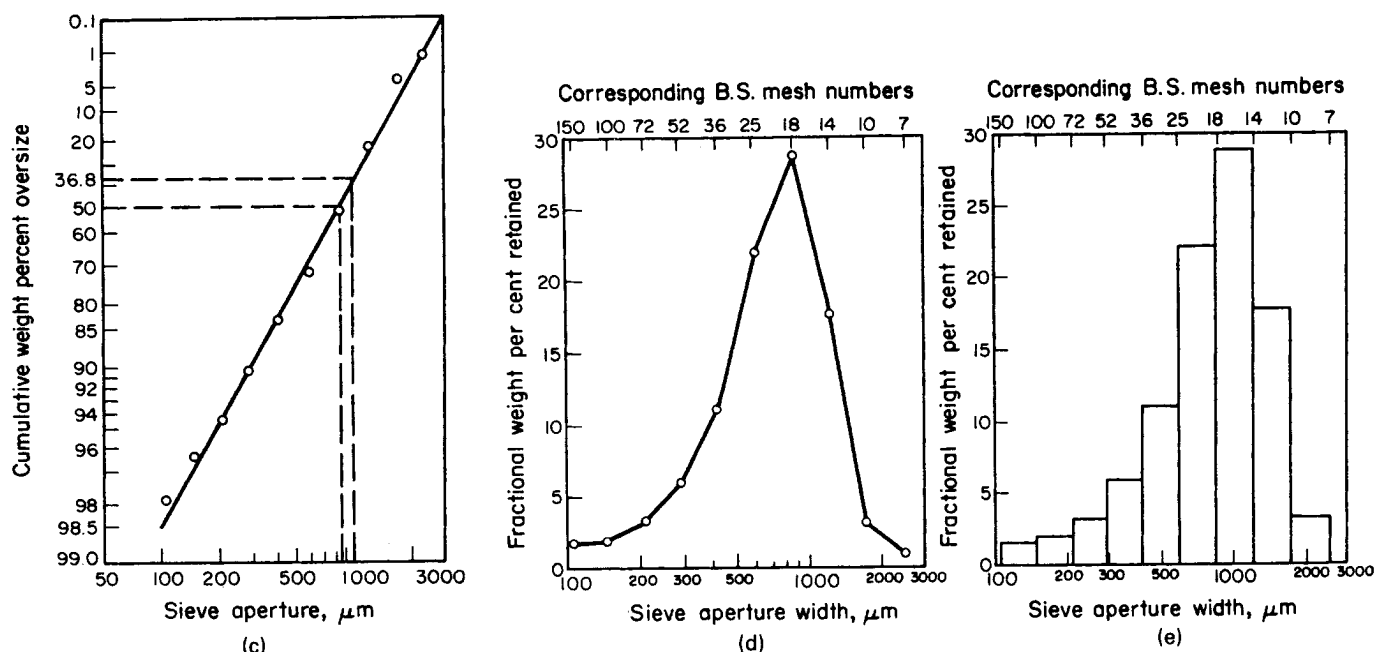


Figure 16.4—(continued)

Nucleation rates are measured by counting the numbers of crystals formed over periods of time.

The growth rates of crystals depend on their instantaneous surface and the linear velocity of solution past the surface as well as the extent of supersaturation, and are thus represented by the

equation

$$\frac{dW_c}{d\theta} = kuA(C - C_0)^n. \quad (16.3')$$

Values of the exponent have been found of the order of 1.5, but

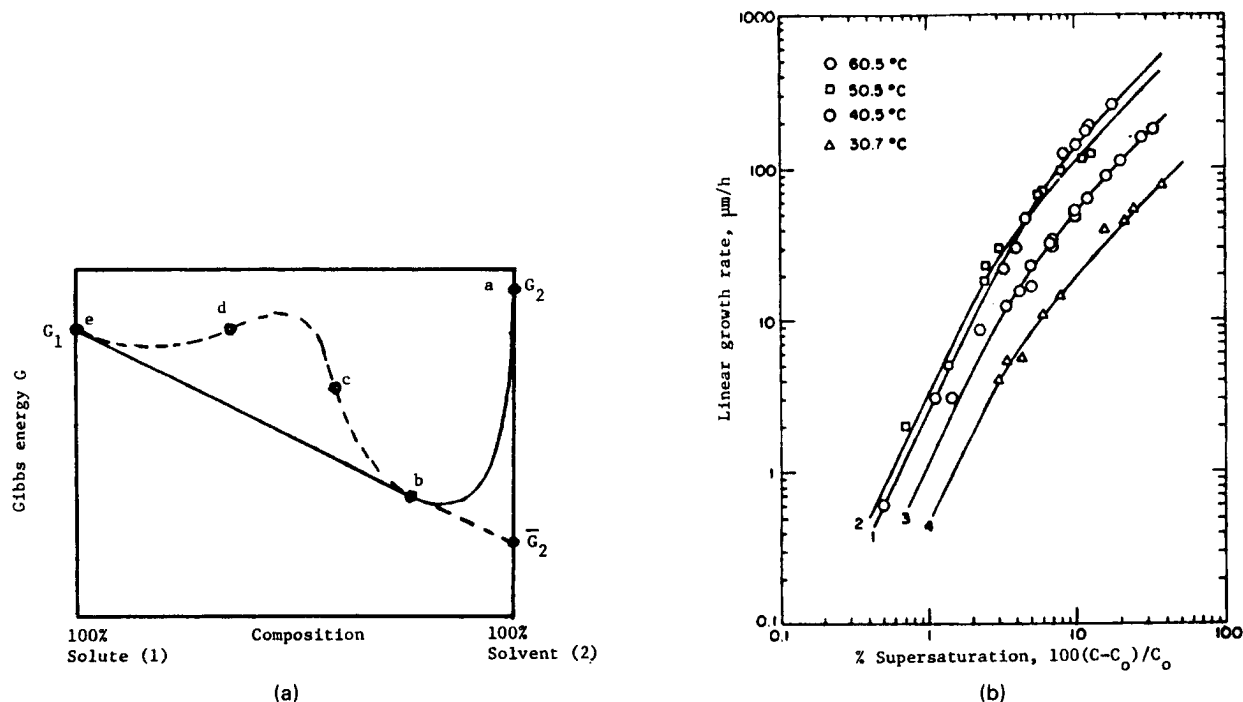


Figure 16.5. Supersaturation behavior. (a) Schematic plot of the Gibbs energy of a solid solute and solvent mixture at a fixed temperature. The true equilibrium compositions are given by points b and e , the limits of metastability by the inflection points c and d . For a salt–water system, point d virtually coincides with the 100% salt point e , with water contents of the order of 10^{-6} mol fraction with common salts. (b) Effects of supersaturation and temperature on the linear growth rate of sucrose crystals [data of Smythe (1967) analyzed by Ohara and Reid, 1973].

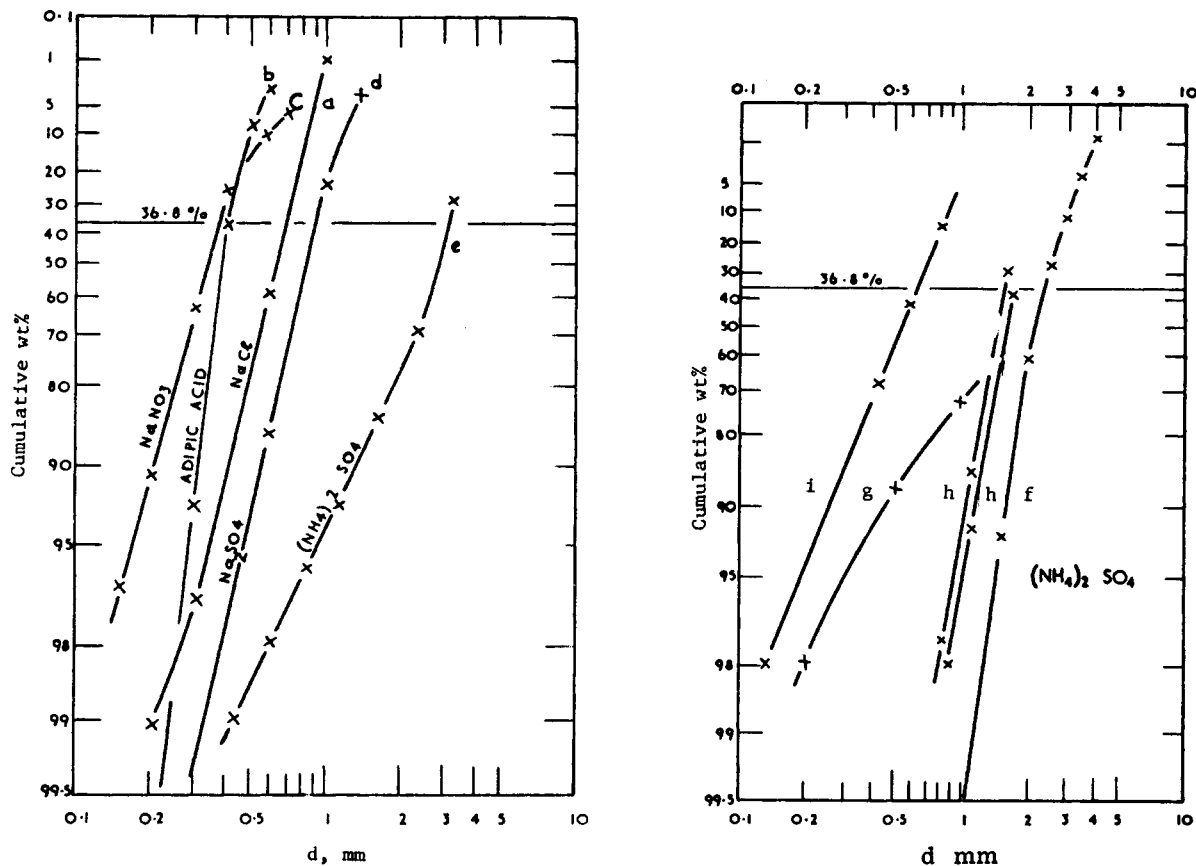


Figure 16.6. Crystal size distributions of several materials in several kinds of crystallizers (Bamforth, 1965).

Code	Crystallizer	Substance	d_m	n
a	Escher-Wyss	NaCl	0.7	4.7
b	Giavanola	adipic acid	0.4	8.1
c	Matusevich	NaNO ₃	0.37	4.0
d	Kestner	Na ₂ SO ₄	0.92	4.7
e	Oslo-Krystal	(NH ₄) ₂ SO ₄	3.2	2.1
f	Oslo-Krystal	(NH ₄) ₂ SO ₄	2.35	6.0
g	Sergeev	(NH ₄) ₂ SO ₄	—	1.5(?)
h	DTB	(NH ₄) ₂ SO ₄	1.6	5.7
i	Standard saturator	(NH ₄) ₂ SO ₄	0.62	2.6

The parameters are those of the RRS equation, Eq. 16.1.

again no correlation of direct use to the design of crystallizers has been achieved. The sucrose growth data of Figure 16.5(b) are not quite log-log linear as predicted by this equation.

In laboratory and commercial crystallizations, wide size distributions usually are the rule, because nuclei continue to form throughout the process, either spontaneously or by breakage of already formed crystals. Large crystals of more or less uniform size are desirable. This condition is favored by operating at relatively low extents of supersaturation at which the nucleation rate is low but the crystals already started can continue to grow. The optimum extent of supersaturation is strictly a matter for direct experimentation in each case. As a rough guide, the data for allowable subcooling and corresponding supersaturation of Table 16.2 may serve. Since the recommended values are one-half the maxima shown, it appears that most crystallizations under

commercial conditions should operate with less than about 2°C subcooling or the corresponding supersaturation. The urea plant design of Example 16.4 is based on 2°F heating.

Growth rates of crystals also must be measured in the laboratory or pilot plant, although the suitable condition may be expressed simply as a residence time. Table 16.3 gives a few growth rate data at several temperatures and several extents of supersaturation for each substance. In most instances the recommended supersaturation measured as the ratio of operating to saturation concentrations is less than 1.1. It may be noted that at a typical rate of increase of diameter of 10^{-7} m/sec, the units used in this table, the time required for an increase of 1 mm is 2.8 hr.

Batch crystallizers often are seeded with small crystals of a known range of sizes. The resulting crystal size distribution for a given overall weight gain can be estimated by an approximate

EXAMPLE 16.4**Deductions from a Differential Distribution Obtained at a Known Residence Time**

The peak of the differential distribution obtained with a residence time of $\bar{t} = 2$ hr corresponds to $L_{pr} = 1.2$ mm. Assuming ideal mixing, $L_{pr}/\bar{G}\bar{t} = 1.2/2G = 3$, and $G = 0.2$ mm/h. With this knowledge of G , crystal size distributions could be found at other residence times.

relation known as the McCabe Delta- L Law, which states that each original crystal grows by the same amount ΔL . The relation between the relative masses of the original and final size distributions is given in terms of the incremental ΔL by

$$R = \frac{\sum w_i (L_{oi} + \Delta L)^3}{\sum w_i L_{oi}^3} \quad (16.4)$$

When R is specified, ΔL is found by trial, and then the size distribution is evaluated. Example 16.5 does this.

Some common substances for which crystallization data are reported in the literature and in patents are listed in Table 16.4.

16.4. THE IDEAL STIRRED TANK

All continuous crystallizers are operated with some degree of mixing, supplied by internal agitators or by pumparound. The important limiting case is that of ideal mixing in which conditions are uniform throughout the vessel and the composition of the effluent is the same as that of the vessel content. In crystallization literature, this model carries the awkward name MSMPR (mixed suspension mixed product removal). By analogy with the terminology of chemical reactors it could be called CSTC (continuous stirred tank crystallizer). Several such tanks in series would be called a CSTC battery. A large number of tanks in series would approach plug flow, but the crystal size distribution still would not be uniform if nucleation continued along the length of the crystallizer.

The process to be analyzed is represented by Figure 16.4. What will be found are equations for the cumulative and differential size distributions in terms of residence time and growth rate. The principal notation is summarized here.

Q = volumetric feed rate,

V_c = volume of holdup in the tank

n = number of crystals per unit volume

L = length of the crystal

G = linear growth rate of the crystal

t = time

$\bar{t} = V_c/Q$, mean residence time

$x = L/\bar{G}\bar{t}$, reduced time

ϕ_m = cumulative mass distribution

n^0 = zero side nuclei concentration, also called zero size population density

B^0 = nucleation rate

a_v = volume shape factor = volume of crystal/(length)³ = $\pi/6$ for spheres, = 1 for cubes.

The case being considered is that in which the feed contains no

TABLE 16.3. Mean Overall Growth Rates of Crystals (m/sec) at Each Face^a

Crystallising substance	°C	S	$\bar{v}$ (m/s)
(NH ₄) ₂ SO ₄ · Al ₂ (SO ₄) ₃ · 24H ₂ O	15	1.03	1.1 × 10 ^{-8*}
	30	1.03	1.3 × 10 ^{-8*}
	30	1.09	1.0 × 10 ^{-7*}
	40	1.08	1.2 × 10 ^{-7*}
NH ₄ NO ₃	40	1.05	8.5 × 10 ⁻⁷
(NH ₄) ₂ SO ₄	30	1.05	2.5 × 10 ^{-7*}
	60	1.05	4.0 × 10 ⁻⁷
	90	1.01	3.0 × 10 ⁻⁸
NH ₄ H ₂ PO ₄	20	1.06	6.5 × 10 ⁻⁸
	30	1.02	3.0 × 10 ⁻⁸
	30	1.05	1.1 × 10 ⁻⁷
	40	1.02	7.0 × 10 ⁻⁸
MgSO ₄ · 7H ₂ O	20	1.02	4.5 × 10 ^{-8*}
	30	1.01	8.0 × 10 ^{-8*}
	30	1.02	1.5 × 10 ^{-7*}
NiSO ₄ · (NH ₄) ₂ SO ₄ · 6H ₂ O	25	1.03	5.2 × 10 ⁻⁹
	25	1.09	2.6 × 10 ⁻⁸
	25	1.20	4.0 × 10 ⁻⁸
K ₂ SO ₄ · Al ₂ (SO ₄) ₃ · 24H ₂ O	15	1.04	1.4 × 10 ^{-8*}
	30	1.04	2.8 × 10 ^{-8*}
	30	1.09	1.4 × 10 ^{-7*}
	40	1.03	5.6 × 10 ^{-8*}
KCl	20	1.02	2.0 × 10 ⁻⁷
	40	1.01	6.0 × 10 ⁻⁷
KNO ₃	20	1.05	4.5 × 10 ⁻⁸
	40	1.05	1.5 × 10 ⁻⁷
K ₂ SO ₄	20	1.09	2.8 × 10 ^{-8*}
	20	1.18	1.4 × 10 ^{-7*}
	30	1.07	4.2 × 10 ^{-8*}
	50	1.06	7.0 × 10 ^{-8*}
	50	1.12	3.2 × 10 ^{-7*}
KH ₂ PO ₄	30	1.07	3.0 × 10 ⁻⁸
	30	1.21	2.9 × 10 ⁻⁷
	40	1.06	5.0 × 10 ⁻⁸
	40	1.18	4.8 × 10 ⁻⁷
NaCl	50	1.002	2.5 × 10 ⁻⁸
	50	1.003	6.5 × 10 ⁻⁸
	70	1.002	9.0 × 10 ⁻⁸
	70	1.003	1.5 × 10 ⁻⁷
Na ₂ S ₂ O ₃ · 5H ₂ O	30	1.02	1.1 × 10 ⁻⁷
	30	1.08	5.0 × 10 ⁻⁷
Citric acid monohydrate	25	1.05	3.0 × 10 ⁻⁸
	30	1.01	1.0 × 10 ⁻⁸
	30	1.05	4.0 × 10 ⁻⁸
Sucrose	30	1.13	1.1 × 10 ^{-8*}
	30	1.27	2.1 × 10 ^{-8*}
	70	1.09	9.5 × 10 ⁻⁸
	70	1.15	1.5 × 10 ⁻⁷

^a The supersaturation is expressed by $S = C/C_0$, with C the amount dissolved and C_0 the normal solubility (kg crystals/kg water). The mean growth velocity is that at one face of the crystal; the length increase is $G = 2\bar{v}$ (m/sec). Data are for crystals in the size range 0.5–1.0 mm in the presence of other crystals. The asterisk denotes that the growth rate probably is size-dependent. (Mullin, 1972).

EXAMPLE 16.5**Batch Crystallization with Seeded Liquor**

Seed crystals with this size distribution are charged to a batch crystallizer:

L_0 , length (mm)	0.251	0.178	0.127	0.089	0.064
w (wt fraction)	0.09	0.26	0.45	0.16	0.04

On the basis of the McCabe ΔL law, these results will be found:

- The length increment that will result in a 20-fold increase in mass of the crystals.
- The mass growth corresponding to the maximum crystal length of 1.0 mm.

When L is the increment in crystal length, the mass ratio is

$$R = \frac{\sum w_i(L_{0i} + L)^3}{\sum w_i L_{0i}^3} = \frac{\sum w_i(L_{0i} + L)^3}{0.09346} = 20$$

- By trial, the value of $L = 0.2804$ mm.
- When $L = 1 - 0.251 = 0.749$, $R = 181.79$.

The size distributions and the computer program are tabulated.

```

10 ! Example 16.4. Batch crystallization with seeded liquor
20 OPTION BASE 1
25 DIM L0(5), W(5), S(5)
30 MAT READ L0, W
40 DATA .251, .178, .127, .089, .064
   .09, .26, .45, .16, .04
50 INPUT L
60 S=0
70 FOR I=1 TO 5
80 S(I)=(L0(I)+L)^3*W(I)
100 S=S(I)+S
110 PRINT USING 120 ; W(I), L0(I), L0(I)+L
120 IMAGE .000, 2X, .000, 2X, 0.0000
130 NEXT I
135 PRINT

```

```

140 PRINT "INCREMENT L="; L
150 PRINT "SUMMATION="; S
160 PRINT "WEIGHT RATIO="; S/.00393458
170 END

```

W	L_0	$L_0 + L$
.090	.251	.2510
.260	.178	.1780
.450	.127	.1270
.160	.089	.0890
.040	.064	.0640

```

INCREMENT L= 0
SUMMATION= .00393458126
WEIGHT RATIO= 1.00000032024

```

.090	.251	.5314
.260	.178	.4584
.450	.127	.4074
.160	.089	.3694
.040	.064	.3444

```

INCREMENT L= .2804
SUMMATION= 7.86768511336E-2
WEIGHT RATIO= 19.9962514763

```

.090	.251	1.0000
.260	.178	.9270
.450	.127	.8760
.160	.089	.8380
.040	.064	.8130

```

INCREMENT L= .749
SUMMATION= .71526668218
WEIGHT RATIO= 181.789843434

```

nuclei but they are generated in the tank. The balance on the number of crystals is

rate of generation = rate of efflux

or

$$V_c \frac{dn}{dt} = Qn \quad (16.5)$$

Upon substituting for the linear growth rate

$$G = dL/dt \quad (16.6)$$

and rearranging,

$$\frac{dn}{n} = \frac{Q}{V_c} dt = \frac{Q}{V_c G} dL = \frac{dL}{iG} = dx \quad (16.7)$$

where

$$\bar{t} = V_c/Q \quad (16.8)$$

is the mean residence time and

$$x = L/G\bar{t} = t/\bar{t} \quad (16.9)$$

is the dimensionless time. Integration of the equation

$$\int_0^n \frac{dn}{n} = \frac{1}{G\bar{t}} \int_0^L dL \quad (16.10)$$

is

$$n = n^0 \exp(-L/G\bar{t}) = n^0 \exp(-x), \quad (16.11)$$

TABLE 16.4. Some Common Substances for which Crystallization Data Are Reported in the Literature and in Patents^a

Compound	Remark or aspect referred to
Ag-halides	
Ag ₂ CrO ₄	growth kinetics
AlF ₃	
Al ₂ O ₃ -corundum	
AlNH ₄ (SO ₄) ₂	
AlK(SO ₄) ₂	influence of supersaturation
Al(OH) ₃	
H ₃ BO ₃	
Na ₂ B ₄ O ₇	oleic acid conducive
BaSO ₄	nucleation growth habit
BaCO ₃	
BaTiO ₄	
CaSO ₄	citrates, SO ₄ ²⁻ , elevated temp.
CaCO ₃	metaphosphate conducive
CaCl ₂	
Ca(NO ₃) ₂	
K ₂ Cr ₂ O ₇	rhythmic crystallisation
CuSO ₄	excess H ₂ SO ₄ detrimental
CuCl ₂	
FeSO ₄	
H ₂ O	nucleation growth
NH ₄ J	nucleation
K-halides	Pb ²⁺ , Zn ²⁺ conducive
KH ₂ PO ₄	
KNO ₃	
K ₂ SO ₄	
K ₂ CrO ₄	
MgSO ₄	<i>t</i> = 45°C, borax conducive
MgCl ₂	
MnCl ₂	
LiF	
LiCl	
Li ₂ SO ₄	
NaCl	Pb, Fe, Al, Zn conducive; caking inhibited by ferrocyanides; urea leads to octahedral prisms
Na ₂ CO ₃	Na ₂ SO ₄ conducive
NaHCO ₃	
Na ₂ SO ₄	wetting agents conducive
Na ₂ S ₂ O ₃	
NaClO ₃	
NaCN	
NH ₄ NO ₃	paraffin, urea, dyes methods of crystallising effect of additives: conducive urea, Fe ²⁺ , Mg ²⁺ , tannin, pH5; Al ³⁺ and Fe ³⁺ lead to needle formation
(NH ₄) ₂ SO ₄	removal of admixtures crystal growth methods of crystallising
(NH ₄) ₂ S ₂ O ₅	
NH ₄ HCO ₃	coarse grained, stabilisation
NH ₄ Cl	Zn ²⁺ , Pb ²⁺ , NH ₄ ⁺ , wood extract
H ₃ PO ₄	
NH ₄ H ₂ PO ₄	Fe ³⁺ and NH ₄ ⁺ conducive
(NH ₄) ₂ HPO ₄	
NiSO ₄	
Pb(NO ₃) ₂	
PbCO ₃	

Compound	Remark or aspect referred to
SrSO ₄	
ZnSO ₄	
anthracene	
adipic acid	
sugars	
citric acid	
phenols	
xylene	
naphthalene	
paraffin	
urea	methods and parameters of crystallisation NH ₄ Cl, MgCO ₃ glyoxal, cyanuric acid surface-active agents
Na-acetate	
NaK-tartrate	
pentaerythrite	
pepsine	
terephthalic acid	

^a(The references, some 400 in number, are given by Nyvlt, 1971 Appendix A).

where

$$n^0 = \lim_{L \rightarrow 0} \frac{dn}{dL} \quad (16.12)$$

is the concentration of crystals of zero length which are the nuclei; it also is called the zero size population density.

The nucleation rate is

$$B^0 = \lim_{L \rightarrow 0} \frac{dn}{dt} = \lim_{L \rightarrow 0} \left(\frac{dL}{dt} \frac{dn}{dL} \right) \quad (16.13)$$

$$= Gn^0. \quad (16.14)$$

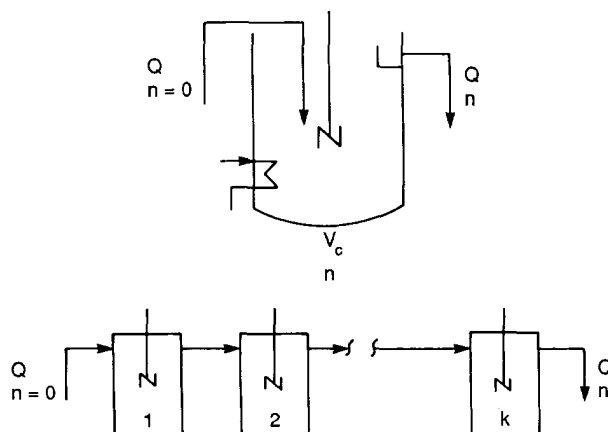


Figure 16.7. Material balancing of continuous stirred tank crystallizers (CSTC). (a) The single stage CSTC. (b) Multistage battery with overall residence time $t = (1/Q) \sum_1^k V_{ci}$.

The number of crystals per unit volume is

$$n_c = \int_0^\infty n dL = \int_0^\infty n^0 \exp(-L/G\bar{t}) dL = n^0 G\bar{t}. \quad (16.15)$$

The total mass of crystals per unit volume is

$$\begin{aligned} m_c &= \int_0^\infty mn dL = \int_0^\infty a_v \rho_c L^3 n^0 \exp(-L/G\bar{t}) dL \\ &= 6a_v \rho_c n^0 (G\bar{t})^4, \end{aligned} \quad (16.16)$$

where a_v is the volumetric shape factor and ρ_c the crystal density. Accordingly, the number of crystals per unit mass is

$$n_c/m_c = 1/6a_v \rho_c (G\bar{t})^3. \quad (16.17)$$

The mass of crystals per unit volume with length less than L or with dimensionless residence time less than x is

$$m_L = \int_0^L mn dL = a_v \rho_c (G\bar{t})^4 n^0 \int_0^x x^3 e^{-x} dx. \quad (16.18)$$

The value of the integral is

$$\int_0^x x^3 e^{-x} dx = 6[1 - e^{-x}(1 + x + x^2/2 + x^3/6)]. \quad (16.19)$$

This expression has a maximum value at $x=3$ and the corresponding length L_{pr} is called the predominant length

$$L_{pr} = 3G\bar{t}. \quad (16.20)$$

The cumulative mass distribution is

$$\phi_m = m_L/m_c = 1 - e^{-x}(1 + x + x^2/2 + x^3/6), \quad (16.21)$$

and the differential mass distribution is

$$d\phi_m/dx = x^3 e^{-x}/6, \quad (16.22)$$

which has a maximum value of 0.224 at $x=3$.

The nucleation rate must generate one nucleus for every crystal present in the product. In terms of M' , the total mass rate of production of crystals,

$$B^0 = \frac{M'}{n_c/m_c} = \frac{M'}{6a_v \rho_c (G\bar{t})^3} = \frac{1.5M'}{a_v \rho_c L_{pr}^3}. \quad (16.23)$$

The principal quantities related by these equations are ϕ_m , $d\phi_m/dx$, L , L_{pr} , $\bar{t}$, n^0 , and B^0 . Fixing a certain number of these will fix the remaining one. Size distribution data from a CSTC are analyzed in Example 16.6. In Example 16.7, the values of the predominant length L_{pr} and the linear growth rate G are fixed. From these values, the residence time and the cumulative and differential mass distributions are found. The effect of some variation in residence time also is found. The values of n^0 and B^0 were found, but they are ends in themselves. Another kind of condition is analyzed in Example 16.4.

MULTIPLE STIRRED TANKS IN SERIES

Operation in several tanks in series will provide narrower size distributions. Equations were developed by Nyvlt (1971) for two main cases. With generation of nuclei in the first stage only, the

cumulative and differential distributions for k stages are

$$\phi_m = 1 - e^{-kx} \sum_{j=0}^{k+2} \frac{n^j x^j}{j!}, \quad (16.24)$$

$$\frac{d\phi_m}{dx} = \frac{k(kx)^{k+2}}{(k+2)!} e^{-kx}. \quad (16.25)$$

The multistage distributions are plotted in Figure 16.8 for several values of the number of stages. Maxima of the differential distributions occur at

$$x_{\max} = 1 + 2/k, \quad (16.26)$$

and the values of those maxima are represented by

$$\left(\frac{d\phi_m}{dx}\right)_{\max} = \frac{k^{k+3}(1+2/k)^{k+2}}{(k+2)!} \exp[-(k+2)]. \quad (16.27)$$

Some numerical values are:

k	1	2	3	4	5	10
$x_{\max}$	3	2	1.67	1.5	1.4	1.2
$(d\phi_m/dx)_{\max}$	0.224	0.391	0.526	0.643	0.745	1.144

Nyvlt (1971) also develops equations for multistage crystallizers in which nuclei form at the same rate in all stages. For two such stages, the cumulative distribution is represented by

$$\begin{aligned} \phi_m &= 1 - 0.5e^{-x}[1 + x + x^2/2 + x^3/6] \\ &\quad - 0.5e^{-2x}[1 + 2x + 2x^2 + (4/3)x^3 + (2/3)x^4]. \end{aligned} \quad (16.28)$$

A comparison of two-stage crystallizers with nucleation in the first stage only and with nucleation in both stages appears in Figure 16.9. The uniformity of crystal size is not as good with nucleation proceeding in every stage; the difference is especially pronounced at larger numbers of stages, which are not shown here but are by Nyvlt (1971).

As in the operation of chemical reactors, multistaging requires shorter residence time for the same performance. For the same L/G ratio, the relative crystallization times of k stages and one stage to reach the peaks are given by Eq. (16.26) as

$$\bar{t}_k/\bar{t}_1 = (1 + k/2)/3, \quad (16.29)$$

which is numerically 0.4 for five stages. Not only is the time shortened, but the size distribution is narrowed. What remains is how to maintain substantial nucleation in only the first stage. This could be done by seeding the first stage and then operating at such low supersaturation that spontaneous nucleation is effectively retarded throughout the battery. Temperature control also may be feasible.

APPLICABILITY OF THE CSTC MODEL

Complete mixing, of course, is not practically realizable and in any event may have a drawback in that intense agitation will cause much secondary nucleation. Some rules for design of agitation of solid suspensions are discussed in Chapter 10, notably in Table 10.2; internal velocities as high as 1.0 ft/sec may be desirable.

Equations can be formulated for many complex patterns, combinations of mixed and plug flow, with decanting of supernatant liquor that contains the smaller crystals and so on. A modification to the CSTC model by Jancic and Garside (1976) recognizes that linear crystal growth rate may be size-dependent; in one instance

EXAMPLE 16.6**Analysis of Size Distribution Data Obtained in a CSTC**

Differential distribution data obtained from a continuous stirred tank crystallizer are tabulated.

w	L	$\Sigma w/L^3$
0.02	0.340	0.5089
0.05	0.430	1.1377
0.06	0.490	1.6477
0.08	0.580	2.0577
0.10	0.700	2.3493
0.13	0.820	2.5851
0.13	1.010	2.7112
0.13	1.160	2.7945
0.10	1.400	2.8310
0.09	1.650	2.8510
0.04	1.980	2.8562
0.03	2.370	2.8584

The last column is of the summation $\Sigma_0^L w_i/L_i^3$ at corresponding values of crystal length L . The volumetric shape factor is $a_v = 0.866$, the density is 1.5 g/mL, and the mean residence time was 2.0 hr. The linear growth rate G and the nucleation rate B^0 will be found.

The number of crystals per unit mass smaller than size L is

$$N = \frac{1}{a_v \rho} \sum_0^L \frac{w_i}{L_i^3}. \quad (1)$$

It is also related to the CSTC material balance by

$$dN/dL = n = n^0 \exp(-L/G\bar{t}). \quad (2)$$

Integration of Eq. (2) is

$$N = \int_0^L n^0 \exp(-L/G\bar{t}) dL = G\bar{t}n^0[1 - \exp(-L/G\bar{t})]. \quad (3)$$

Combining Eqs. (1) and (3),

$$\sum w_i/L_i^3 = a_v \rho G\bar{t}n^0[1 - \exp(-L/G\bar{t})]. \quad (4)$$

The two unknowns G and n^0 may be found by nonlinear regression with the 12 available data for L_i . However, two representative

values of L_i are taken here, and the unknowns are solved for by simultaneous solution of two equations. When

$$L = 0.58, \quad \Sigma = 2.0577,$$

$$L = 1.40, \quad \Sigma = 2.8310.$$

Substituting into Eq. (4) and ratioing,

$$\frac{2.8310}{2.0577} = \frac{1 - \exp(-1.4/G\bar{t})}{1 - \exp(-0.58/G\bar{t})};$$

by trial,

$$G\bar{t} = 0.5082$$

$$G = 0.5082/2 = 0.2541.$$

With $L = 1.4$ in Eq. (4),

$$2.8310 = 0.866(1.5)(0.5082)n^0[1 - \exp(-1.4/0.5082)],$$

from which

$$\begin{aligned} n^0 &= 4.58 \text{ nuclei/mm}^4 \\ &= 4.58(10)^{12} \text{ nuclei/m}^4. \end{aligned}$$

Accordingly,

$$B^0 = Gn^0 = 0.2541(10)^{-3}(4.58)(10)^{12} = 1.16(10)^9 \text{ nuclei/m}^3 \text{ hr}.$$

The cumulative mass size distribution is represented by

$$\phi_m = 1 - e^{-x}(1 + x + x^2/2 + x^3/6)$$

with

$$x = L/G\bar{t} = L/0.5082.$$

This distribution should be equivalent to the original one, but may not check closely because the two points selected may not have been entirely representative. Moreover, although the data were purportedly obtained in a CSTC, the mixing may not have been close to ideal.

they find that

$$G = G^0(1 + L/G^0\bar{t})^{0.65}.$$

Other studies have tried to relate sizes of draft tubes, locations and sizes of baffles, circulation rate, and so on to crystallization behavior. So far the conclusions are not general enough to do a designer much good. A possibly useful concept, the separation index (SI), is mentioned by Mullin (1976, p. 293):

$$SI = (\text{kg of 1 mm equivalent crystals})/\text{m}^3 \text{ hr}.$$

For inorganic salts in water at near ambient temperature, a value of SI in the range of 100–150 kg/m³/hr may be expected. An illustration of the utilization of pilot plant data and plant experience in the design of a urea crystallizer is in Example 16.1.

In general, the design policy to be followed is to utilize as much

laboratory and pilot plant information as possible, to work it into whatever theoretical pattern is applicable, and to finish off with a comfortable safety factor. There may be people who know how; they should be consulted.

16.5. KINDS OF CRYSTALLIZERS

The main kinds of crystallizers are represented in Figure 16.10. They will be commented on in order. Purification of products of melt crystallization is treated separately.

Batch crystallizers are used primarily for production of fine chemicals and pharmaceuticals at the rate of 1–100 tons/week. The one exception is the sugar industry that still employs batch vacuum crystallization on a very large scale. In that industry, the syrup is concentrated in triple- or quadruple-effect evaporators, and crystallization is completed in batch vacuum pans that may or may not be equipped with stirrers [Fig. 16.11(g)].

EXAMPLE 16.7**Crystallization in a Continuous Stirred Tank with Specified Predominant Crystal Size**

Crystals of citric acid monohydrate are to be made in a CSTC at 30°C with predominant size $L_{pr} = 0.833$ mm (20 mesh). The density is 1.54 g/mL, the shape factor $a_v = 1$ and the solubility is 39.0 wt %. A supersaturation ratio $C/C_0 = 1.05$ is to be used.

Take the growth rate, $G = 2\bar{v}$, to be one-half of the value given in Table 16.3:

$$G = dL/d\theta = 4(10^{-8}) \text{ m/sec, } 0.144 \text{ mm/hr.}$$

The predominant size is related to other quantities by

$$L_{pr} = 0.833 = 3G\bar{t},$$

from which

$$\bar{t} = 0.833/(3)(0.144) = 1.93 \text{ hr.}$$

For a mass production rate of 15 kg/hr of crystals, $C = 15$, the nucleation rate is

$$B^0 = \frac{1.5C}{a_v \rho_c L_{pr}^3} = \frac{1.5(15)}{1(1.5)[0.833(E-3)]^3} \\ = 2.595(10)^{10} \text{ nuclei/m}^3 \text{ hr.}$$

The zero size concentration of nuclei is

$$n^0 = B^0/G = 2.595(10)^{10}/4(10)^{-8} = 6.49(10)^{17} \text{ nuclei/m}^4.$$

Accordingly, the equation of the population density is

$$n = n^0 \exp(-L/G\bar{t}) = \exp(41.01 - 360L).$$

The cumulative mass distribution is

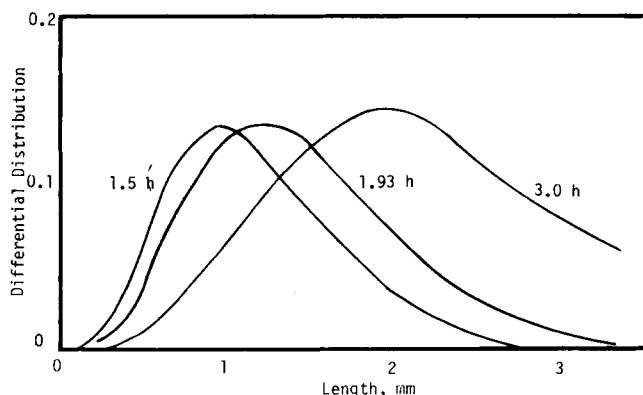
$$\phi_m = 1 - e^{-x}(1 + x + x^2/2 + x^3/6),$$

where

$$x = L/G\bar{t} = 3.60L, \text{ with } L \text{ in mm.}$$

The differential distributions are differences between values of ϕ_m at successive values of crystal length L . The tabulation shows cumulative and differential distributions at the key $\bar{t} = 1.93$ hr, and also at 1.5 and 3.0 hr. The differential distributions are plotted and show the shift to larger sizes as residence time is increased, but the heights of the peaks are little affected.

Mesh	mm	$\bar{t} = 1.5 \text{ h}$		$\bar{t} = 1.93 \text{ h}$		$\bar{t} = 3.0 \text{ h}$	
		Cum	Diff	Cum	Diff	Cum	Diff
0	0	1.0000	.0002	1.0000	.0020	1.0000	.0517
6	3.327	.9998	.0010	.9980	.0068	.9483	.0623
7	2.794	.9989	.0041	.9912	.0185	.8859	.0913
8	2.362	.9948	.0136	.9728	.0424	.7947	.1226
9	1.981	.9812	.0350	.9304	.0778	.6720	.1410
10	1.651	.9462	.0603	.8526	.1021	.5310	.1260
12	1.397	.8859	.0983	.7505	.1322	.4051	.1183
14	1.168	.7876	.1152	.6183	.1278	.2868	.0873
16	.991	.6724	.1343	.4905	.1268	.1995	.0693
20	.833	.5381	.1304	.3637	.1071	.1302	.0482
24	.701	.4076	.1157	.2565	.0845	.0820	.0323
28	.589	.2919	.0929	.1720	.0614	.0497	.0204
32	.495	.1990	.0684	.1106	.0416	.0293	.0123
35	.417	.1306	.0483	.0690	.0274	.0169	.0074
42	.351	.0823	.0324	.0416	.0173	.0096	.0043
48	.295	.0499	.0312	.0243	.0108	.0053	.0025
60	.246	.0287	.0119	.0135	.0058	.0028	.0013
65	.208	.0168	.0073	.0077	.0034	.0015	.0007
80	.175	.0095	.0043	.0042	.0019	.0008	.0004
100	.147	.0052	.0037	.0023	.0016	.0004	.0003
150	.104	.0015	.0011	.0006	.0005	.0001	.0001
200	.074	.0004	.0004	.0002	.0002	.0000	.0000



Natural circulation evaporators like those shown on Figure 8.16 may be equipped for continuous salt removal and thus adapted to crystallization service. For large production rates, however, forced circulation types such as the DTB crystallizer of Figure 16.10(g), with some control of crystal size, are the most often used. The lower limit for economic continuous operation is 1–4 tons/day of crystals, and the upper limit in a single vessel is 100–300 tons/day, but units in parallel can be used for unlimited capacity.

Many special types of equipment have been developed for particular industries, possibly extreme examples being the simple open ponds for solar evaporation of brines and recovery of salt, and the specialized vacuum pans of the sugar industry that operate with syrup on the tubeside of calandrias and elaborate internals to eliminate entrainment. Some modifications of basic types of crystallizers often carry the inventor's or manufacturer's name. For their identification, the book of Bamforth (1965) may be consulted.

The basic equipment descriptions following carry the letter designations of Figure 16.10.

(a) Jacketed pipe scraped crystallizers. These are made with inner pipe 6–12 in. dia and 20–40 ft long, often arranged in tiers of three or more connected in series. Scraper blades rotate at 15–30 rpm. Temperatures of -75 to $+100^\circ\text{F}$ have been used and viscosities in excess of 10,000 cP present no problems. Although the action is plug flow with tendency to uniform crystal size, the larger particles settle to the bottom and grow at the expense of the smaller ones that remain suspended, with the result that a wide range of sizes is made. Capacity is limited by rates of heat transfer; coefficients of 10–25 Btu/(hr)(sqft)($^\circ\text{F}$) usually are attainable. Higher coefficients are obtainable in Votators (Cherry Burrell Co.) that have more intense scraping action. Pilot units of 4 in. by 4 ft and larger are made.

(b) Swenson–Walker type. In comparison with jacketed pipes, they have the advantage of being more accessible for cleaning. The standard unit is 24 in. wide, 26 in. high, and 10 ft long. Four units in line may be driven off one shaft. Capacity is limited by heat transfer rates which may be in the range of 10–25 Btu/(hr)(sqft)($^\circ\text{F}$), with an

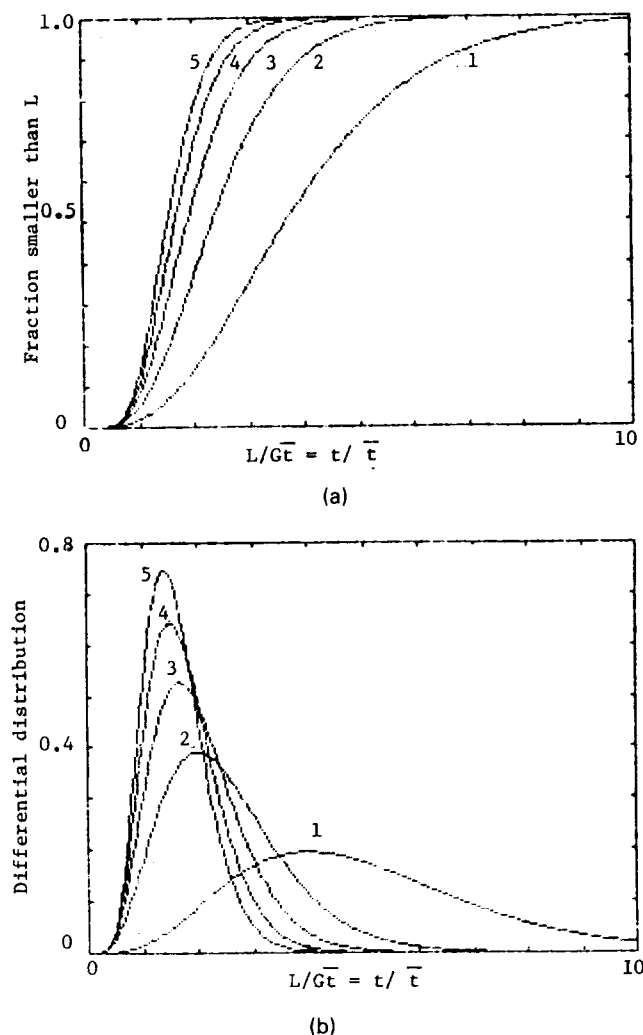


Figure 16.8. Theoretical crystal size distributions from an ideal stirred tank and from a series of tanks with generation of nuclei only in the first tank. Equations of the curves and for the peak values are in the text. (a) Cumulative distributions. (b) Differential distributions.

effective area of 3 sqft/ft of length. According to data in *Chemical Engineers' Handbook* (3rd ed., McGraw-Hill, New York, 1950, p. 1071), a 40 ft unit is able to produce 15 tons/day of trisodium phosphate, and a 50 ft unit can make 8 tons/day of Glaubers salt. The remarks about crystal size distribution made under item (a) apply here also.

(c) Batch stirred and cooled types. Without agitation, crystallization time can be 2–4 days; an example is given in *Chemical Engineer's Handbook* (1950, p. 1062). With agitation, times of 2–8 hr are sometimes cited. The limitation is due to attainable rates of heat transfer. Without encrustation of surfaces by crystals, coefficients of 50–200 Btu/(hr)(sqft)(°F) are realizable, but temperature differences are maintained as low as 5–10°F in order to keep supersaturation at a level that prevents overnucleation. Stirring breaks corners off crystals and results in secondary nucleation so that crystal size is smaller than in unagitated tanks. Larger crystal sizes are obtained by the standard practice of seeding with an appropriate range of fine crystals. Calculation of the performance of such an operation is made in

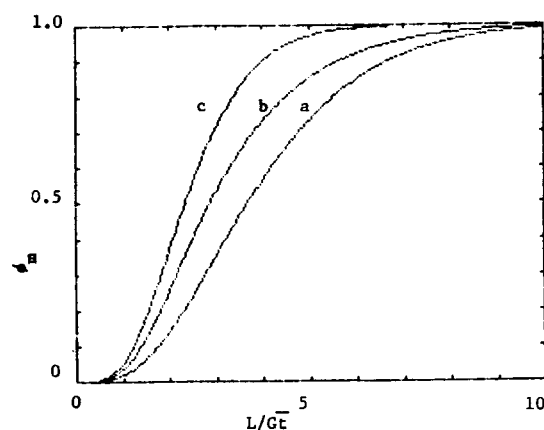


Figure 16.9. Cumulative size distribution in continuous stirred tanks. (a) one tank; (b) two tanks in series, nucleation in both; (c) two tanks in series, nucleation in only the first.

Example 16.5. Teflon heat transfer tubes that are thin enough to flex under the influence of circulating liquid cause a continual descaling that maintains good heat transfer consistently, 20–65 Btu/(hr)(sqft)(°F). Circulating types such as Figures (d) and (e) often are operated in batch mode, the former under vacuum if needed. High labor costs keep application of batch crystallizers to small or specialty production.

(d) Circulating evaporators. Some units are built with internal coils or calandrias and are simply conventional evaporators with provisions for continual removal of crystals. Forced circulation and external heat exchangers provide better temperature control. High velocities in the tubes keep the surfaces scoured. Temperature rise is limited to 3–10°F per pass in order to control supersaturation and nucleation. Operation under vacuum often is practiced. When the boiling point elevation is not excessive, the off vapors may be recompressed and used again for heating purposes. Multiple effect units in series for thermal economy may be used for crystallizing evaporators as they are for conventional evaporation. Pilot units of 2 ft dia are made, and commercial units up to 40 ft dia or so.

(e) Circulating cooling crystallizers. Such operations are feasible when the solubility falls sharply with decreasing temperature. Coolers usually are applied to smaller production rates than the evaporative types. Cooling is 1–2°F per pass and temperature differences across the tubes are 5–15°F.

The special designs of Figure 16.11 mostly feature some control of crystal size. They are discussed in order.

(a) Draft tube baffle (DTB) crystallizer. The growing crystals are circulated from the bottom to the boiling surface with a slow moving propeller. Fine crystals are withdrawn from an annular space, redissolved by heating to destroy unwanted nuclei and returned with the feed liquor. The temperature rise caused by mixing of heated feed and circulating slurry is 1–2°F. The fluidized bed of large crystals at the bottom occupies 25–50% of the vessel volume. Holdup time is kept sufficient for crystal growth to the desired size. Products such as KCl, $(\text{NH}_4)_2\text{SO}_4$, and $(\text{NH}_4)_2\text{H}_2\text{PO}_4$ can be made in this equipment in the range of 6–20 mesh. Reaction and crystallization can be accomplished simultaneously in DTB units. The reactants can be charged into the recirculation line or into the draft tube. Examples are the production of ammonium sulfate from ammonia and sulfuric acid and the neutralization of waste acids with lime. The heat of reaction is removed by evaporation of water.

(b) Direct contact refrigeration. Such equipment is operated as

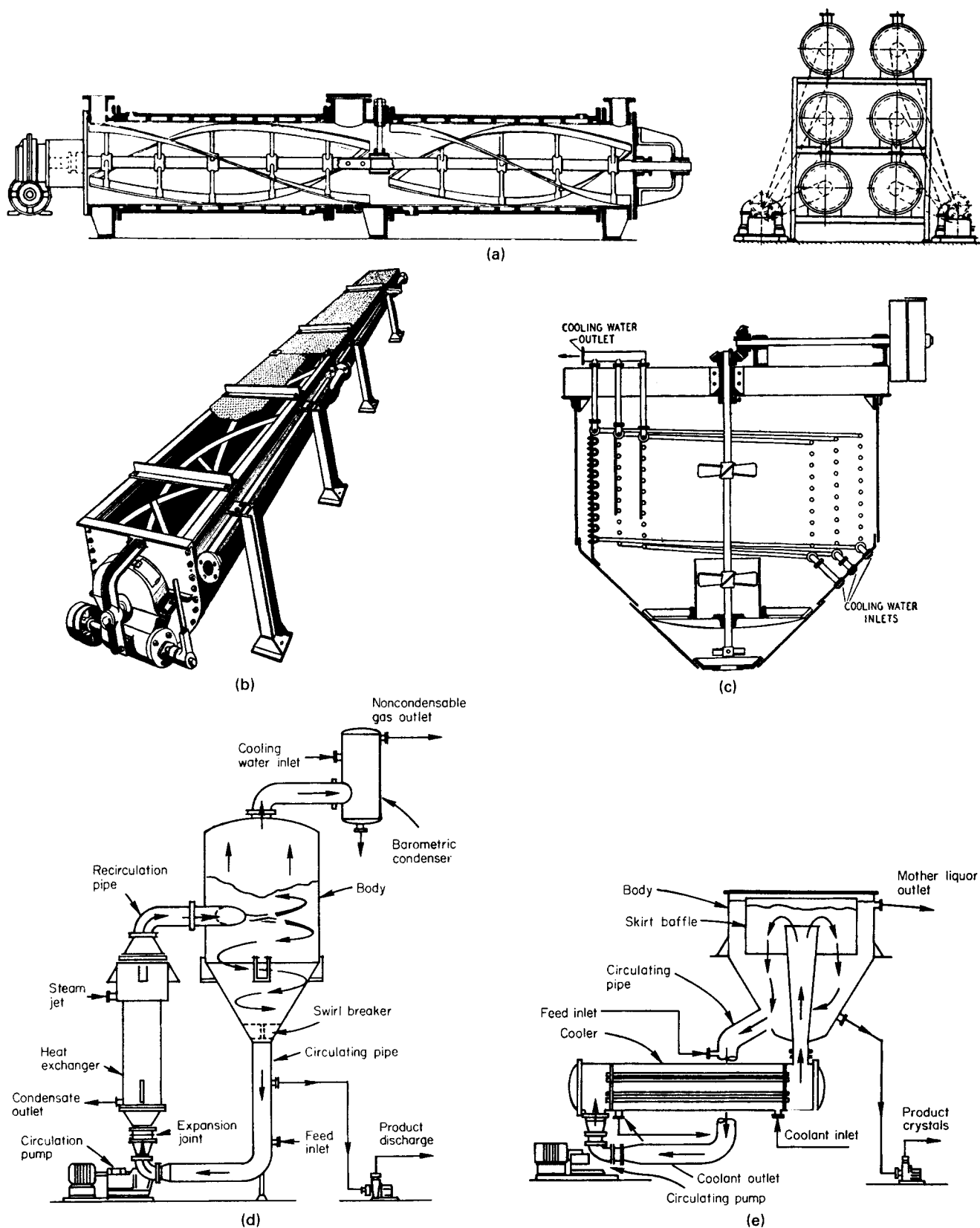


Figure 16.10. Basic types of batch and continuous crystallizers. (a) Jacketed scraped pipe and assembly of six units (Riegel, *Chemical Process Machinery*, Reinhold, N.Y., 1953). (b) Swenson-Walker jacketed scraped trough (Swenson Evaporator Co., Riegel, 1953). (c) Batch stirred tank with internal cooling coil (Badger, and McCabe, *Elements of Chemical Engineering*, McGraw-Hill, New York, 1936). (d) Crystallization by evaporation, with circulation through an external heater (Schweitzer, loc. cit., p. 2.170). (e) Crystallization by chilling, with circulation through an external cooler. (P.A. Schweitzer, Ed., *Handbook of Separation Techniques for Chemical Engineers*, McGraw-Hill, New York, 1979, p. 2.166).

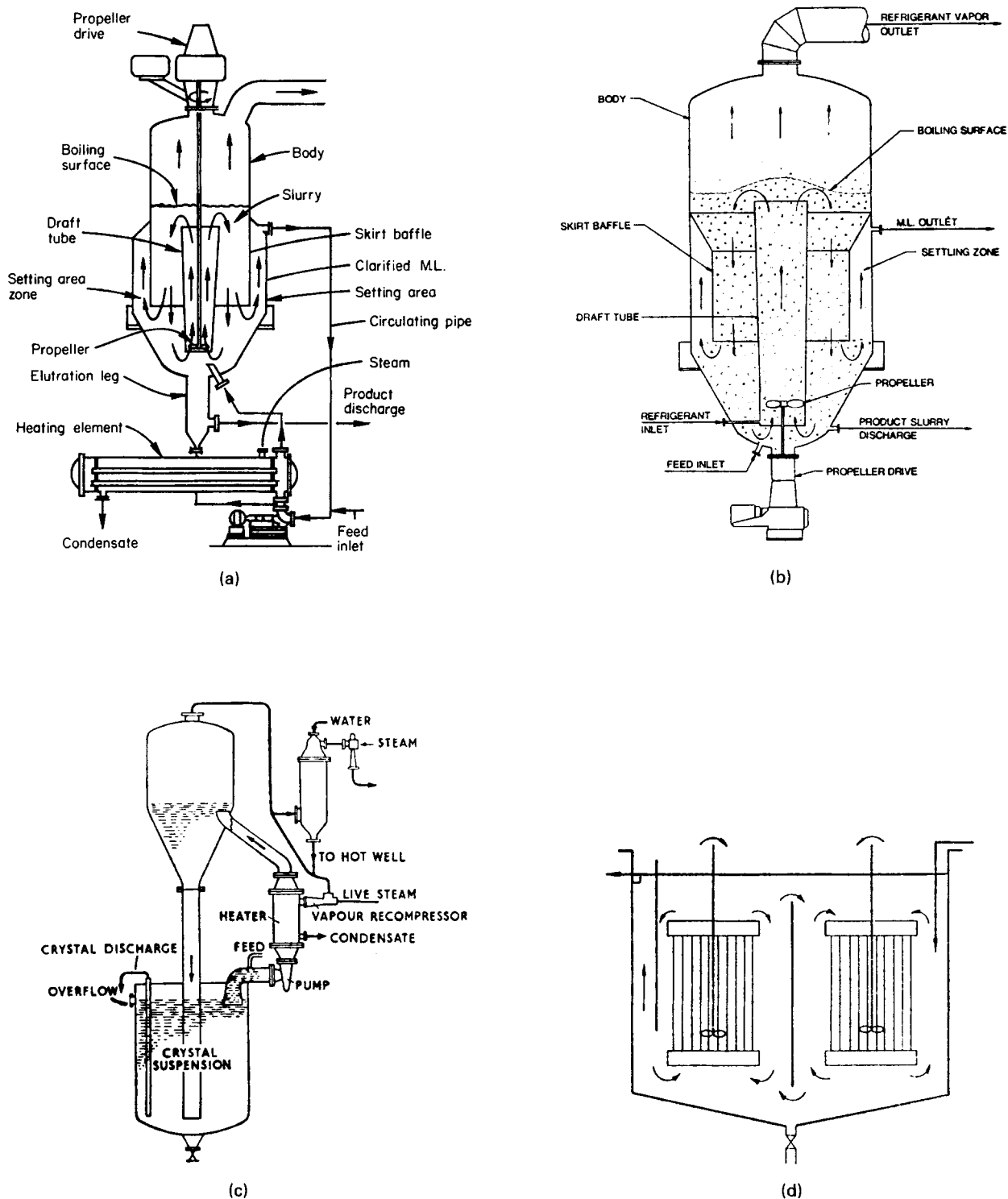
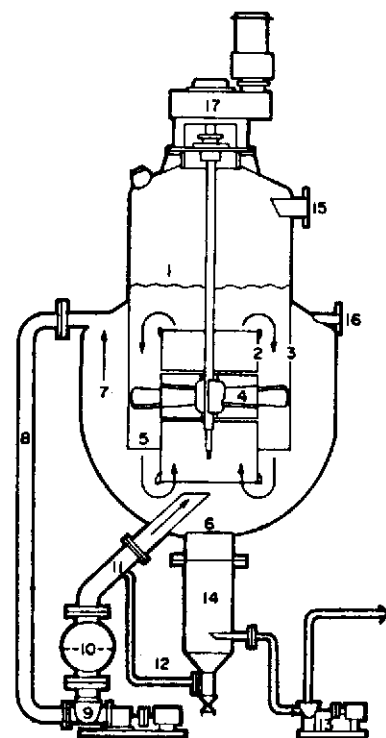
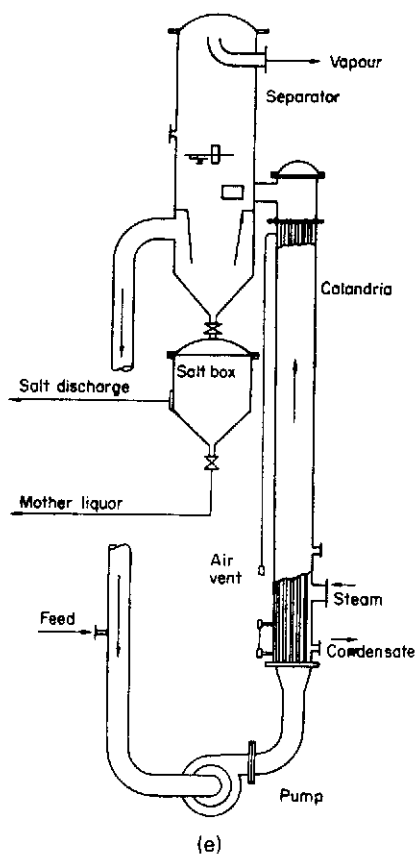
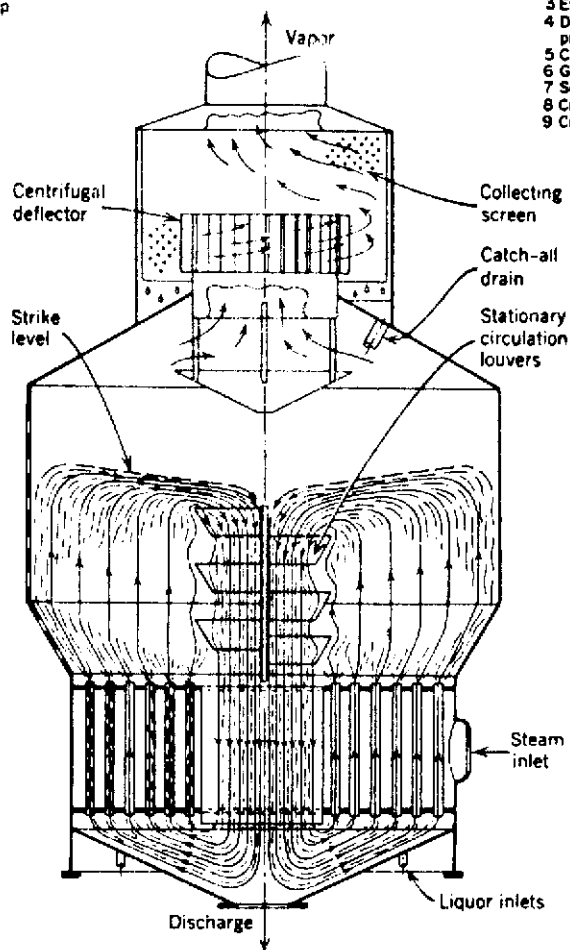


Figure 16.11. Examples of special kinds of crystallizers. (a) Swenson draft tube baffle (DTB) crystallizer; crystals are brought to the surface where growth is most rapid, the baffle permits separation of unwanted fine crystals, resulting in control of size. (b) Direct chilling by contact with immiscible refrigerant, attains very low temperatures and avoids encrustation of heat transfer surfaces. Freons and propane are in common use. (c) Oslo "Krystal" evaporative classifying crystallizer. Circulation is off the top, the fine crystals are destroyed by heating, large crystals grow in the body of the vessel. (d) Twinned crystallizer. When one chamber is maintained slightly supersaturated and the other slightly subsaturated, coarse crystals can be made. (Nyvlt, 1971). (e) APV-Kestner long tube salting evaporator; large crystals (0.5 mm or so) settle out. (f) Escher-Wyss or Tsukushima DP (double propeller) crystallizer. The double propeller maintains upward flow in the draft tube and downward flow in the annulus, resulting in highly stable suspensions. (g) A vacuum pan for crystallization of sugar (Honolulu Iron Works).



- | | |
|---------------------------------------|----------------------------|
| 1 Thickening zone | 10 Heater or cooler |
| 2 Draft tube | 11 Solution return |
| 3 Evaporation chamber | 12 Elutriation liquid feed |
| 4 Double-acting circulation propeller | 13 Slurry discharge pump |
| 5 Crystal growth zone | 14 Elutriation zone |
| 6 Grading zone | 15 Vapour outlet |
| 7 Settling zone | 16 Overflow |
| 8 Circulation pipe | 17 Variable-speed drive |
| 9 Circulation pump | |

(f)



(g)

Figure 16.11—(continued)

low as -75°F . Essentially immiscible refrigerant is mixed with the liquor and cools it by evaporation. The effluent refrigerant is recovered, recompressed, and recycled. Direct contacting eliminates the need for temperature difference across a heat transfer tube which can be economically more than $5\text{--}15^{\circ}\text{F}$, and also avoids scaling problems since the liquor must be on the outside of the tubes when refrigerant is used. Examples are crystallization of caustic with freon or propane and of *p*-xylene with propane refrigerant.

(c) Oslo "Krystal" evaporative classifying crystallizer. The supernatant liquid containing the fines is circulated through the external heater where some of the fines are redissolved because of the temperature rise. The settled large crystals are withdrawn at the bottom. The recirculation rate is much greater than the fresh feed rate. In one operation of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystallization, fresh feed saturated at 120°C is charged at 2000 kg/hr to the vessel maintained at 40°C and is mixed with a recirculated rate of $50,000\text{ kg/hr}$ to produce a mixture that is temporarily at 43°C , which then evaporates and cools. Vessel sizes as large as 15 ft dia and 20 ft high are mentioned in the literature. The same principle is employed with cooling type crystallization operations.

(d) Twinned crystallizer. Feed is to the right chamber. The rates of recirculation and forward feed are regulated by the position of the center baffle. Improved degree of uniformity of crystal size is achieved by operating one zone above saturation temperature and the other below. Fine particles are dissolved and the larger ones grow at their expense. Even with both zones at the same temperature, the series operation of two units in series gives more nearly uniform crystal size distribution than can be made in a single stirred tank. It is not stated if any such crystallizers are operated outside Nyvlt's native land, Czechoslovakia, that also produces very fine tennis players (Lendl, Mandlikova, Navratilova, Smid, and Sukova).

(e) APV-Kestner long tube vertical evaporative crystallizers are used to make small crystals, generally less than 0.5 mm , of a variety of substances such as NaCl , Na_2SO_4 , citric acid, and others; fine crystals recirculate through the pump and heater.

(f) Escher-Wyss (Tsukushima) double propeller maintains flow through the draft tube and then annulus and maintains highly stable suspension characteristics.

(g) Sugar vacuum pan. This is an example of the highly specialized designs developed in some long-established industries. Preconcentration is effected in multiple effect evaporators; then crystallization is accomplished in the pans.

16.6. MELT CRYSTALLIZATION AND PURIFICATION

Some mixtures of organic substances may be separated advantageously by cooling and partial crystallization. The extent of such recovery is limited by the occurrence of eutectic behavior. Examples 16.2 and 16.8 consider such limitations. Sometimes these limitations can be circumvented by additions of other substances that change the phase equilibria or may form easily separated compounds with one of the constituents that are subsequently decomposed for recovery of its constituents.

Thus the addition of *n*-pentane to mixtures of *p*-xylene and *m*-xylene permits complete separation of the xylenes which form a binary eutectic with 11.8% para. Without the *n*-pentane, much para is lost in the eutectic, and none of the meta is recoverable in pure form. A detailed description of this process is given by Dale (1981), who calls it extractive crystallization. Other separation processes depend on the formation of high melting molecular compounds or clathrates with one of the constituents of the mixture. One example is carbon tetrachloride that forms a compound with *p*-xylene and alters the equilibrium so that its separation from *m*-xylene is

facilitated. Hydrocarbons form high-melting hydrates with water; application of propane hydrate formation for the desalination of water has been considered. Urea forms crystalline complexes with straight chain paraffins such as the waxy ingredients of lubricating oils. After separation, the complex may be decomposed at $75\text{--}80^{\circ}\text{C}$ for recovery of its constituents. This process also is described by Dale (1981). Similarly thiourea forms crystalline complexes with isoparaffins and some cyclic compounds.

Production rates of melt crystallization of organic materials usually are low enough to warrant the use of scraped surface crystallizers like that of Figure 16.10(a). A major difficulty in the production of crystals is the occlusion of residual liquor on them which cuts the overall purity of the product, especially so because of low temperatures near the eutectic and the consequent high viscosities. Completeness of removal of occluded liquor by centrifugation or filtration often is limited because of the fragility and fineness of the organic crystals.

MULTISTAGE PROCESSING

In order to obtain higher purity, the first product can be remelted and recrystallized, usually at much higher temperatures than the eutectic so that occlusion will be less, and of course at higher concentration. In the plant of Figure 16.12, for instance, occlusion from the first stage is 22% with a content of 8% *p*-xylene and an overall purity of 80% ; from the second stage, occlusion is 9% with a PX content of 42% but the overall purity is 95% PX; one more crystallization could bring the overall purity above 98% or so.

Because the handling of solids is difficult, particularly that of soft organic crystals, several crystallization processes have been developed in which solids do not appear outside the crystallizing equipment, and the product leaves the equipment in molten form. For organic substances, crystalline form and size usually are not of great importance as for products of crystallization from aqueous solutions. If needed, the molten products can be converted into flakes or sprayed powder, or in extreme cases they can be recrystallized out of a solvent.

THE METALLWERK BUCHS PROCESS

The Metallwerk Buchs (MWB) process is an example of a batch crystallization that makes a molten product and can be adapted to multistaging when high purities are needed. Only liquids are transferred between stages; no filters or centrifuges are needed. As appears on Figure 16.13, the basic equipment is a vertical thin film shell-and-tube heat exchanger. In the first phase, liquor is recirculated through the tubes as a film and crystals gradually freeze out on the cooled surface. After an appropriate thickness of solid has accumulated, the recirculation is stopped. Then the solid is melted and taken off as product or transferred to a second stage for recrystallization to higher purity.

PURIFICATION PROCESSES

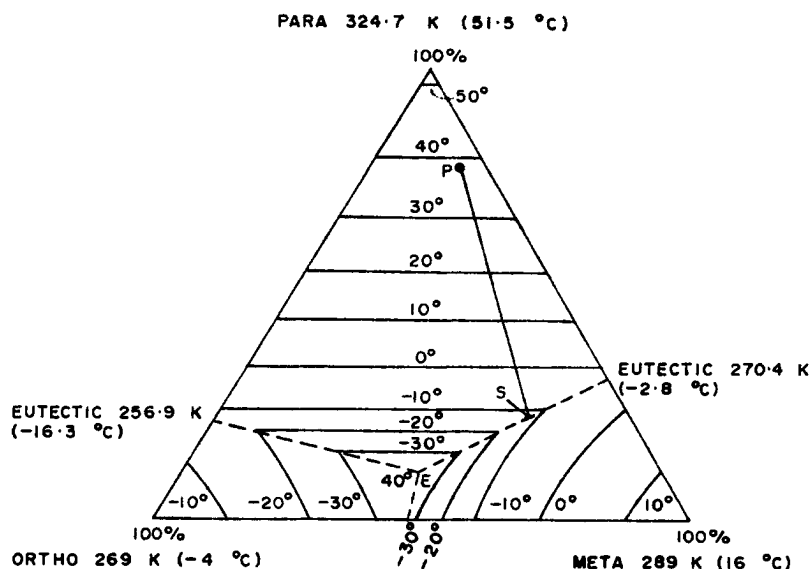
As an alternative to multistage batch crystallization processes with their attendant problems of material handling and losses, several types of continuous column crystallizers have been developed, in which the product crystals are washed with their own melts in countercurrent flow. Those illustrated in Figures 16.14–16.17 will be described. Capacities of column purifiers as high as 500 gal/(hr) (sqft) have been reported but they can be less than one-tenth as much. Lengths of laboratory size purifiers usually are less than three feet.

Schildknecht Column [Fig. 16.14(a)]. This employs a rotating spiral or screw to move the solids in the direction against the flow of

EXAMPLE 16.8**Crystallization from a Ternary Mixture**

The case is that of mixtures of the three isomeric nitrotoluenes for which the equilibrium diagram is shown. Point *P* on the diagram has the composition 0.885 para, 0.085 meta, and 0.030 ortho. The temperature at which crystals begin to form must be found experimentally or it may be calculated quite closely from the heats and temperatures of fusion by a method described for instance by Walas (Example 8.9, *Phase Equilibria in Chemical Engineering*, Butterworths, Stoneham, MA, 1985). It cannot be found with the data shown on the diagram. In the present case, incipient freezing is at 46°C, with para coming out at point *P* on the diagram. As cooling

continues, more and more pure para crystals form. The path is along straight line *PS* which corresponds to constant proportions of the other two isomers since they remain in the liquid phase. At point *S*, -13°C, which is on the eutectic trough of meta and ortho, the meta also begins to precipitate. Para and meta continue to precipitate along the trough until the ternary eutectic *E* is reached at -40°C when complete solidification occurs. The cooling path is shown on the phase diagram. The recovery of pure para at equilibrium at various temperatures and the composition of the liquid phase are tabulated. (Coulson and Warner, *A Problem in Chemical Engineering Design: The Manufacture of Mononitrotoluene*, Inst. of Chem. Eng., Rugby, England, 1949).



Temperature, °C	Para- deposited, kg	Mother liquor, kg	Composition of mother liquor (per cent by weight)		
			O	M	P
46	0	100	3.0	8.5	88.5
40	39.6	60.4	5.0	14.0	81.0
30	66.7	33.3	8.7	24.8	66.5
20	75.0	25.0	12.0	34.0	54.0
10	79.6	20.4	14.7	41.6	43.7
0	82.3	17.7	16.7	48.0	35.3
-10	84.8	15.2	17.0	53.9	27.1

the fluid. The conveyor is of open construction so that the liquid can flow through it but the openings are small enough to carry the solids. Throughputs of 50 L/hr have been obtained in a 50 mm dia column. Because of the close dimensional tolerances that are needed, however, columns larger than 200 mm dia have not been successful. Figure 16.14(a) shows a section for the formation of the crystals, but columns often are used only as purifiers with feed of crystals from some external source.

Philips Crystallization Process [Fig. 16.14(b)]. The purifying equipment consists of a vessel with a wall filter and a heater at the

bottom. Crystals are charged from an external crystallizer and forced downwards with a reciprocating piston or with pulses from a pump. The washing liquid reflux flows from the melting zone where it is formed upward through the crystal bed and out through the wall filter. Pulse displacement is 0.3–0.6 cm/sqcm of column cross section, with a frequency of 200–250/min. For many applications reflux ratios of 0.05–0.60 are suitable. Evaluation of the proper combination of reflux and length of purifier must be made empirically.

From a feed containing 65% *p*-xylene, a column 1000 sqcm in cross section can make 99% PX at the rate of 550 kg/hr, and 99.8%

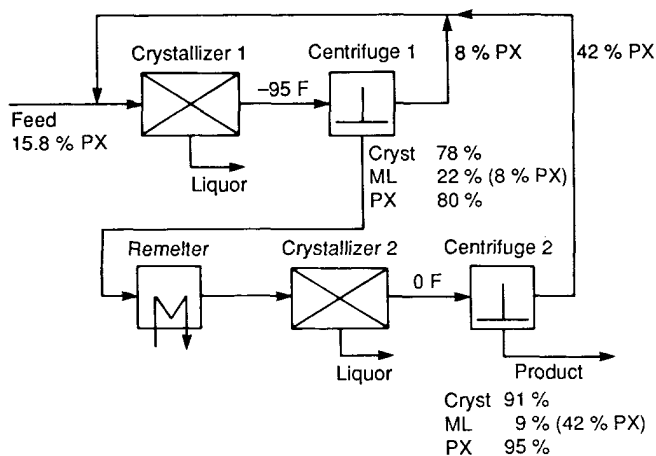


Figure 16.12. Humble two-stage process for recovery of *p*-xylene by crystallization. Yield is 82.5% of theoretical. ML = mother liquor, PX = *p*-xylene (Haines, Powers and Bennett, 1955).

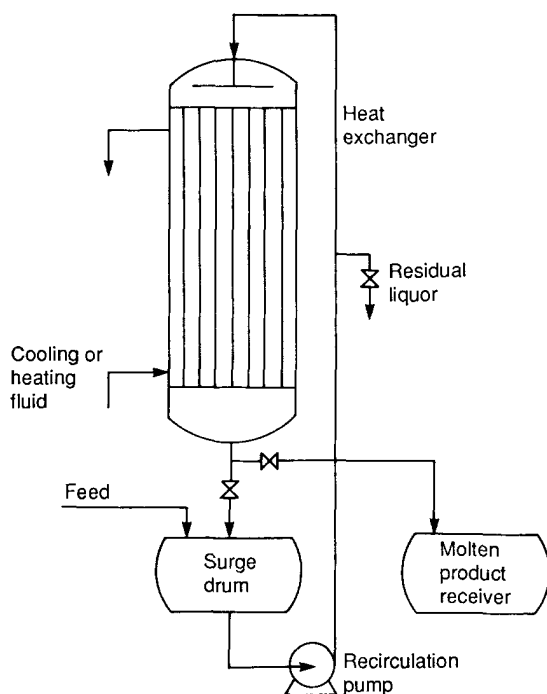


Figure 16.13. MWB (Metallwerk Buchs) batch recirculating crystallizer, with freezing on and melting off insides of thin film heat exchanger tubes; adaptable to multistage processing without external solids handling (Mützenberg and Saxer, 1971).

PX at 100 kg/hr; this process has been made obsolete, however, by continuous adsorption with molecular sieves. Similarly, a feed of 83 mol % of 2-methyl-5-vinyl pyridine has been purified to 95% at the rate of 550 g/hr cm² and 99.7% at 155 g/hr cm². At one time, columns of more than 60 cm dia were in operation.

Brodie Crystallizer-Purifier [Fig. 16.14(c)]. This equipment combines a horizontal scraped surface crystallizer with a vertical purifying section. The capacity and performance of the purifier

depends strongly on the sizes of the crystals that enter that zone. In order to ensure adequate crystal size, residence times in the crystallizing zone as long as 24 hr may be needed. No data of residence times are stated in the original article. Some operating data on the recovery of para-dichlorobenzene from a mixture containing 75% of this material are reported for a purifier that is 1.14 sqft cross section as follows, as well as data for some other materials.

Reflux ratio	2	0.5	0.25
Feed rate (gal/hr)	29	60	90
Residue rate (gal/hr)		20	30
Product rate (gal/hr)	20	40	60
PDCB in residue (%)		25	25
Product purity (%)	99.997	99.99	99.5

TNO Bouncing Ball Purifier (Fig. 16.15). The basis for this design is the observation that small crystals melt more readily and have a greater solubility than large ones. The purifier is a column with a number of sieve trays attached to a central shaft that oscillates up and down. As the slurry flows through the tower, bouncing balls on each tray impact the crystals and break up some of them. The resulting small crystals melt and enrich the liquid phase, thus providing an upward refluxing action on the large crystals that continue downward to the melting zone at the bottom. Reflux is returned from the melting zone and product is taken off.

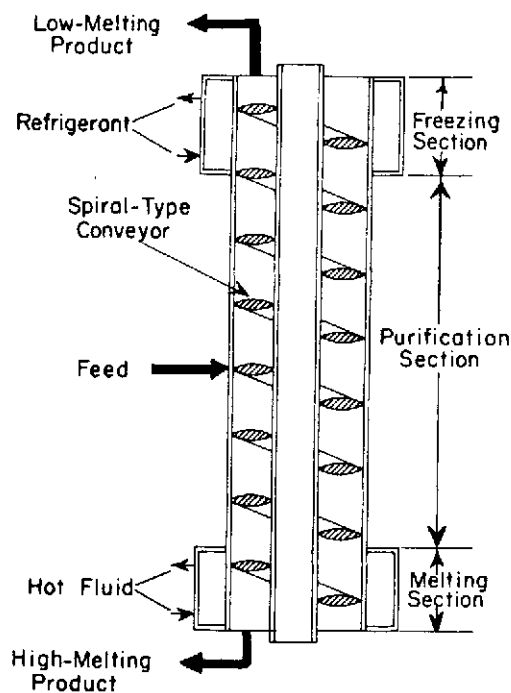
Specifications of a pilot plant column are:

diameter, 80 mm,
hole size, 0.6 × 0.6 mm,
number of balls/tray, 30,
diameter of balls, 12 mm,
amplitude of vibration, 0.3 mm,
frequency, 50/sec,
number of trays, 13,
tray spacing, 100 mm.

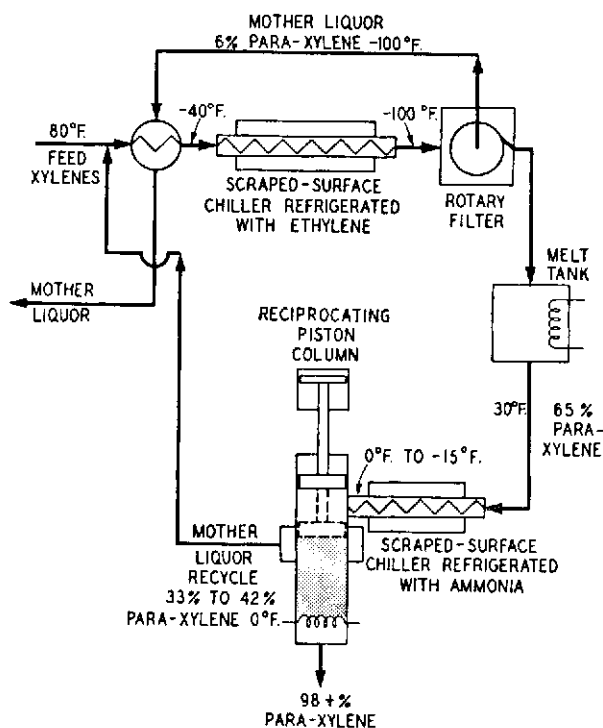
For the separation of benzene and thiophene that form a solid solution, a tray efficiency of more than 40% could be realized. Flow rates of 100–1000 kg/m² hr have been tested. The residence time of crystals was about 30 min per stage. Eutectic systems also have been handled satisfactorily. A column 500 mm dia and 3 m long with 19 trays has been built; it is expected to have a capacity of 300 tons/yr.

Kureha Double-Screw Purifier (Fig. 16.16). This unit employs a double screw with intermeshing blades that express the liquid from the crystal mass as it is conveyed upward. The melt is formed at the top, washes the rising crystals countercurrently, and leaves as residue at the bottom. A commercial unit has an effective height of 2.6 m and a cross section of 0.31 m². When recovering 99.97% *p*-dichlorobenzene from an 87% feed, the capacity is 7000 metric tons/yr. The feed stock comes from a tank crystallizer and filter. Data on other eutectic systems are shown, and also on separation of naphthalene and thiophene that form a solid solution; a purity of 99.87% naphthalene is obtained in this equipment.

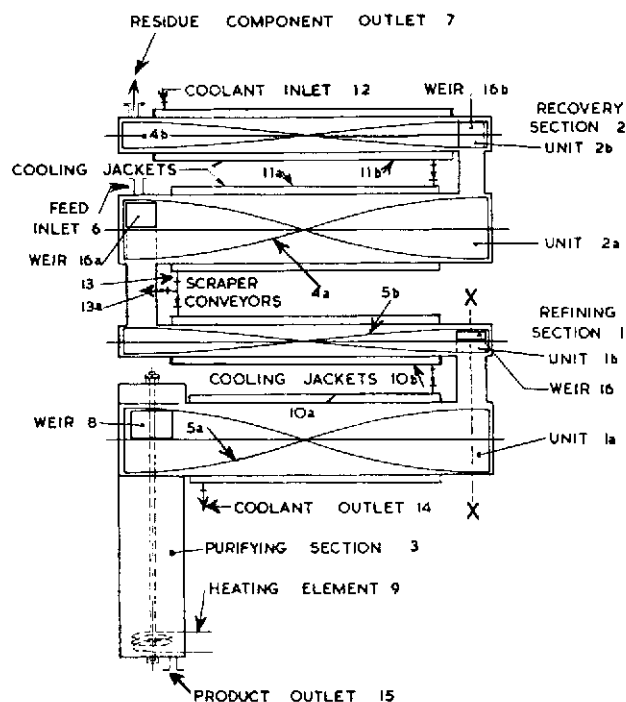
Brennan-Koppers Purifier (Fig. 16.17). This equipment employs top melting like the Kureha and wall filters like the Philips. Upward movement of the crystals is caused by drag of the flowing fluid. The crystal bed is held compact with a rotating top plate or piston that is called a harvester. It has a corrugated surface that scrapes off the top of the bed and openings that permit the crystals to enter the melting zone at any desired rate. The melt flows downward through the openings in the harvester, washes the



(a)



(b)



(c)

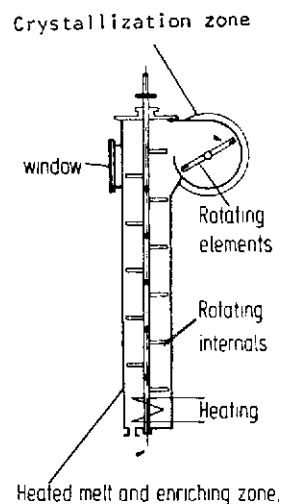


Figure 16.14. Three types of crystal purifiers with different ways of transporting the crystals. (a) Spiral or screw conveyor type, laboratory scale, but successful up to 200 mm dia [Schildknecht, *Z. Anal. Chem.* **181**, 254 (1961)]. (b) Philips purifier with reciprocating piston or pulse pump drive [McKay, Dale, and Weedman, *Ind. Eng. Chem.* **52**, 197 (1960)]. (c) Combined crystallizer and purifier, gravity flow of the crystals; purifier details on the right (Brodie, 1971).

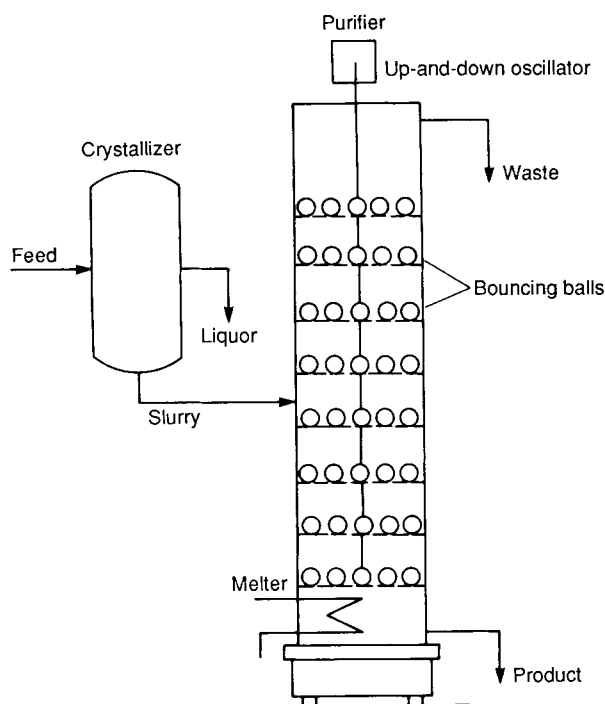
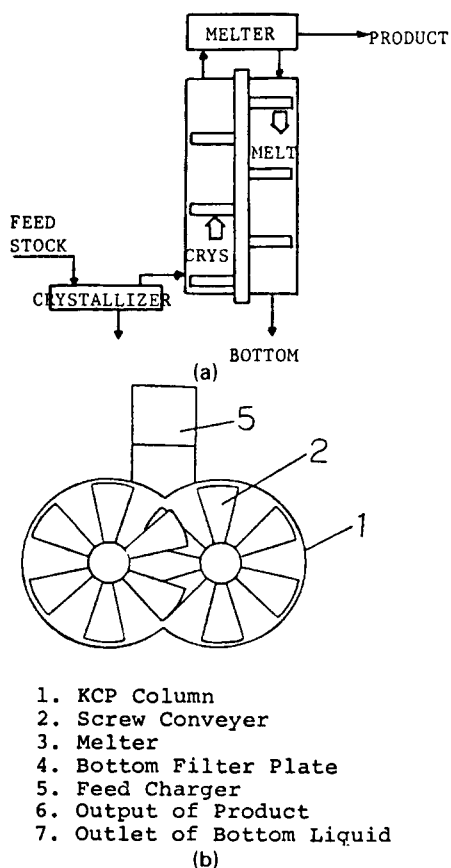


Figure 16.15. TNO Crystal Purifier (Arkenbout *et al.*, 1976, 1978).

upwardly moving crystals, and leaves through the sidewall filter as residue. The movement of crystals is quite positive and not as dependent on particle size as in some other kinds of purifiers. Data are given in the patent (U.S. Pat. 4,309,878) about purification of 2,6-ditertiary butyl para cresol; the harvester was operated at 40–60 rpm and filtration rates of 100 lb/(hr)(sqft) were obtained. Other information supplied directly by E.D. Brennan are that a 24 in. dia unit stands 9 ft high without the mixer and that the following performances have been achieved:

	Diameter (in.)	Purity (wt %)		Prod. Rate (lb/hr/ft ²)
		Feed	Product	
A. Pilot plant tests				
Acetic acid	3	83	99.85	100
<i>p</i> -Dichlorobenzene	6	70	99.6	380
Naphthalene (high sulfur)	6	68	98	220
Di- <i>t</i> -butyl- <i>p</i> -cresol	3	85	99.1	210
	6	85	99.1	230
B. Commercial operation				
Di- <i>t</i> -butyl- <i>p</i> -cresol	24	90	99.5	340

All feeds were prepared in Armstrong scraped surface crystallizers.



1. KCP Column
2. Screw Conveyor
3. Melter
4. Bottom Filter Plate
5. Feed Charger
6. Output of Product
7. Outlet of Bottom Liquid

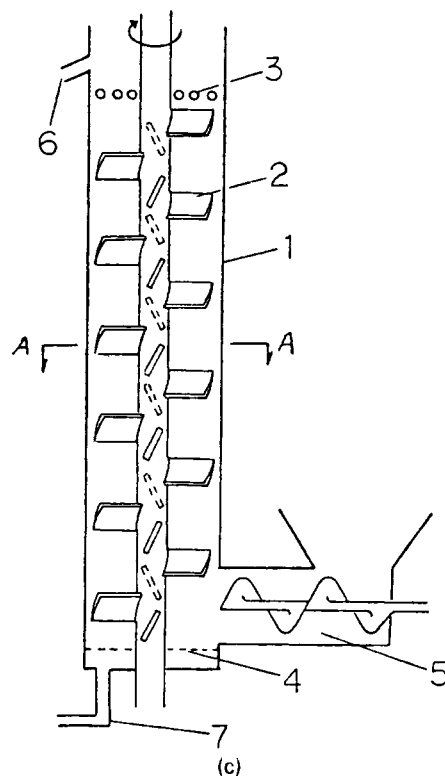


Figure 16.16. Kureha continuous crystal purifier (KCP column) (Yamada, Shimizu, and Saitoh, in Jancic and DeJong, 1982, pp. 265–270). (a) Flowsketch. (b) Dumbbell-shaped cross section at AA. (c) Details of column and screw conveyor.

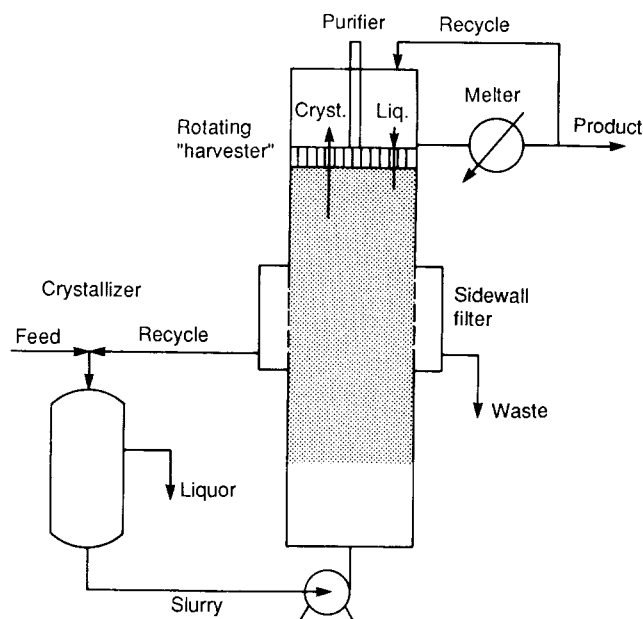


Figure 16.17. Brennan-Koppers crystal purifier (Brennan, 1982).

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